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Identifying and Rewriting Between Portuguese Varieties with Fine-tuned Language Models

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Declaration
I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

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Abstract

Portuguese has multiple varieties, but language models are shaped by the predominance of Brazilian Portuguese in the data used for training. This work studies European Portuguese (EP) and Brazilian Portuguese (BP) through two tasks: variety identification and bidirectional rewriting.

The models use data from OpenSubtitles, Few-shot Region-aware Machine Translation (FRMT), the Golden Collection, and GPT-Wikipedia, a resource created from pairs focused only on differences between the varieties and processed to reduce unnecessary changes.

We use 270M- and 4B-parameter T5Gemma 2 encoder-decoder models. The 270M model is fully fine-tuned to select the training protocol, while the 4B model uses Low-Rank Adaptation (LoRA) to compare control placement, data composition, and three-stage training.

Data composition produces the largest differences in rewriting quality. Models trained only with GPT-Wikipedia achieve better results on the Golden Collection, where sentences require few changes. Adding FRMT improves identification on that dataset and reduces performance on the Golden Collection, with rewriting effects depending on direction and control placement. Decoder control performs better in every BP-to-EP comparison, while EP-to-BP results are mixed. Reinforcement Learning continuation changes the GPT-Wikipedia models only slightly but substantially recovers Golden Collection performance after GPT-Wikipedia + FRMT. The results also show that automatic metrics do not always distinguish necessary changes from appropriate preservation or broader reformulations.

Keywords

Portuguese language varieties; European Portuguese; Brazilian Portuguese; Variety identification; Text rewriting; Model fine-tuning

Resumo

A língua Portuguesa apresenta múltiplas variedades, mas os modelos de linguagem são condicionados pela predominância do Português do Brasil nos dados de treino. Este trabalho estuda o Português Europeu (EP) e o Português do Brasil (BP) em duas tarefas: identificação da variedade e reescrita bidirecional.

Os dados usados nos modelos provêm do OpenSubtitles, do Few-shot Region-aware Machine Translation (FRMT), da Golden Collection e do GPT-Wikipedia, criado com pares centrados apenas nas diferenças entre variedades e cujo processamento reduz alterações desnecessárias.

Usam-se modelos T5Gemma 2 encoder-decoder de 270M e 4B parâmetros. O modelo de 270M recebe fine-tuning completo para selecionar o protocolo de treino, e o de 4B usa Low-Rank Adaptation (LoRA) para comparar a posição do controlo, a composição dos dados e o treino em três etapas.

A composição dos dados produz as maiores diferenças na qualidade da reescrita. Modelos treinados apenas com GPT-Wikipedia obtêm melhores resultados na Golden Collection, onde as frases requerem poucas alterações. Adicionar FRMT melhora a identificação nesse dataset e reduz o desempenho na Golden Collection, com efeitos na reescrita dependentes da direção e do controlo. O controlo no decoder é melhor em todas as comparações BP-para-EP, enquanto EP-para-BP apresenta resultados mistos. A continuação baseada em Reinforcement Learning altera pouco os modelos GPT-Wikipedia, mas recupera substancialmente o desempenho na Golden Collection após GPT-Wikipedia + FRMT. Os resultados também mostram que as métricas automáticas nem sempre distinguem alterações necessárias de preservação adequada ou reformulações amplas.

Palavras Chave

Variedades da língua portuguesa; Português europeu; Português brasileiro; Identificação de variedade; Reescrita textual; Fine-tuning de modelos

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Acronyms

ALMA	Advanced Language Model-based trAnslator
BERT	Bidirectional Encoder Representations from Transformers
BLEU	Bilingual Evaluation Understudy
BP	Brazilian Portuguese
DPO	Direct Preference Optimization
DSL-TL	Discriminating Between Similar Languages – True Labels
EP	European Portuguese
FRMT	Few-shot Region-aware Machine Translation
GRPO	Group Relative Policy Optimization
ID	Identifier
KL	Kullback–Leibler
LLM	Large Language Model
LoRA	Low-Rank Adaptation
LSTM	Long Short-Term Memory
MT	Machine Translation
NLP	Natural Language Processing
NMT	Neural Machine Translation
PEFT	Parameter-Efficient Fine-Tuning
POS	Part of Speech
PPO	Proximal Policy Optimization
QE	Quality Estimation
QLoRA	Quantized Low-Rank Adaptation

RL	Reinforcement Learning
RLHF	Reinforcement Learning from Human Feedback
sBLEU	sentence-level BLEU
SFT	Supervised Fine-Tuning
TER	Translation Edit Rate
UL2	Unifying Language Learning Paradigms
URL	Uniform Resource Locator
XLM-R	XLM-RoBERTa

1

Introduction

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This chapter introduces the problem addressed in this work and frames the scope of the work. It begins by motivating Portuguese variety processing from both linguistic and data-centric perspectives (Section 1.1). It then states the research objective guiding this work (Section 1.2), summarizes the main contributions (Section 1.3), and describes the organization of the remaining chapters (Section 1.4).[1]

1.1 Motivation

Large language models (LLMs) are increasingly used for multilingual text processing, including tasks such as classification, generation, translation, and structured text transformation (Zhao et al., 2026). Even so, this success does not change the basic limitation that their behavior remains shaped by the distribution of the data on which they were trained. For languages with multiple standardized varieties, this creates a practical problem. Comparable challenges arise in languages such as English, Spanish, Arabic, or French, where users may expect systems to respect their variety rather than the one most represented in training data (Zampieri et al., 2020).

Brazilian Portuguese is disproportionately represented in Portuguese text available on the web, and resources labeled simply as Portuguese tend to overrepresent Brazilian usage. That imbalance can

[1] Code and reproducibility artifacts: GitHub. Reported model weights: Hugging Face. Curated data and provenance: Hugging Face.

propagate to pretrained models, which may treat Brazilian Portuguese as the unmarked default and under-serve European Portuguese users (Sousa et al., 2025b).

Portuguese is a pluricentric language shared across countries and speech communities whose varieties have been shaped by geographic separation, migration, institutional standardization, media ecosystems, and local cultural practices. Among these varieties, European Portuguese (EP) and Brazilian Portuguese (BP) are studied in this work as two national standards with differences in spelling, lexical choice, discourse conventions, and morphosyntactic preferences such as clitic placement, verbal periphrases, subject realization, or forms of address (Barreiro and Mota, 2018). Therefore, rewriting between EP and BP often has to remain conservative, with many sentences requiring only localized lexical or morphosyntactic changes and others remaining unchanged.

Computationally, these differences are systematic enough to affect model outputs, while remaining easy to ignore during data collection, annotation, and evaluation. As a result, corpora labeled only at the language level often mix multiple varieties, and models trained on them can blur distinctions that remain important to users. Automatic filtering can further affect this representation when corpus-construction pipelines depend on language-identification predictions. Under domain mismatch, class imbalance, and language similarity, identification errors can propagate into web corpora (Caswell et al., 2020).

1.2 Research Statement

Research on Natural Language Processing (NLP) for language varieties has grown in recent years, but work that simultaneously studies EP/BP identification, EP/BP rewriting, and adaptation under curated supervision remains comparatively scarce. Portuguese is used in this work as a case study for that broader problem. Rather than treating Portuguese as a single homogeneous language, this work studies how corpus construction, task formulation, supervision composition, control placement, and adaptation strategy influence variety-aware behavior.

Accordingly, the central objective of this work is to determine how pretrained encoder-decoder models can be adapted to identify and rewrite EP/BP varieties under data constraints. More specifically, it aims to determine how preprocessing, filtering, and corpus composition affect the supervision available for training and evaluation, and to investigate how task and variety information should be represented within a shared encoder-decoder interface. In the rewriting task, textual control strings indicate the requested operation and the variety of the input sentence, which determines the rewriting direction. The study compares placing control information on the encoder side with the variety label on the decoder side.

The experimental study uses T5Gemma 2 models with 270M and 4B parameters for different experimental roles. The 270M-parameter model is fully fine-tuned for lower-cost supervised protocol selection and separate reward-design diagnostics, whereas the 4B-parameter model is adapted with Low-Rank Adaptation (LoRA) (Hu et al., 2022) for the control, data-composition, and staged-training experiments. To investigate these questions, the experiments with the 4B-parameter model follow a three-stage adaptation pipeline. Stage A provides broad supervised EP/BP exposure through OpenSubtitles, Stage B performs targeted specialization with GPT-Wikipedia either alone or with additional Few-shot Region-aware

Machine Translation (FRMT) supervision, and Stage C applies reward-based continuation from the corresponding Stage B checkpoints using more conservative rewriting-only subsets. The resulting models are evaluated on the Golden Collection (Sanches et al., 2024) and FRMT (Riley et al., 2023), providing evaluation settings for analyzing variety identification and rewriting behavior.

1.3 Contributions

Accordingly, this research project makes four main contributions.

1. It consolidates reusable EP-BP data resources from multiple sources. Sources include OpenSubtitles (Tiedemann, 2009), the Golden Collection (Sanches et al., 2024), FRMT (Riley et al., 2023), and an additional GPT-Wikipedia resource. The associated preprocessing and filtering procedures are adapted to the characteristics of each source.

2. It develops and evaluates models for both language variety identification and bidirectional rewriting between EP and BP.

3. It studies 270M T5Gemma 2 full fine-tuning for supervised protocol selection and 4B LoRA for control placement, data composition, staged training, and reward-based continuation.

4. It complements aggregate evaluation with corpus and qualitative error analyses, highlighting data strengths, model failures, and linguistic phenomena that remain difficult after fine-tuning.

These contributions are connected through the data and modeling choices made in this work. On the data side, corpus construction makes heterogeneous resources comparable. OpenSubtitles, the Golden Collection, FRMT, and GPT-Wikipedia differ in origin, format, noise profile, alignment quality, and supervision type. For that reason, this work includes preprocessing, filtering, and task-rendering steps that convert these materials into shared formats for EP/BP identification and rewriting, rather than treating them as directly interchangeable inputs.

On the modeling side, the contribution is a study of T5Gemma 2 (Zhang et al., 2025b) encoder-decoder models with 270 million and 4 billion parameters. The 270M-parameter model is fully fine-tuned to compare alternative supervised-training protocols before the experiments with the 4B-parameter model. Evaluation uses accuracy and macro-F1 for identification, and Bilingual Evaluation Understudy (BLEU) (Papineni et al., 2002) and Translation Edit Rate (TER) (Snover et al., 2006) for rewriting. In addition to these quantitative metrics, the analysis examines copy behavior, conservative edits, paraphrasing, and over-generation, which are important in this setting where the preferred output should remain close to the source and change only what is necessary. This makes the evaluation sensitive to data-quality problems that may be hidden by corpus-level scores alone.

The experiments built from these contributions point to a conditional pattern. Among the supervised comparisons, data composition produces the largest rewriting shifts. GPT-Wikipedia aligns most closely with conservative Golden Collection rewriting, whereas the GPT-Wikipedia + FRMT data mix encourages broader changes and lowers Golden Collection performance. The effects of control placement and

staged adaptation vary across resources and rewriting directions. Reward-based continuation changes the GPT-Wikipedia conditions modestly while partly recovering Golden Collection performance in the GPT-Wikipedia + FRMT conditions. The copy analysis shows that overlap metrics reward restraint but cannot distinguish necessary variety adaptation, acceptable paraphrase, and unnecessary reformulation.

1.4 Document Organization

The remainder of this work is organized as follows. Chapter 2 introduces the linguistic and technical background required to understand the problem and the modeling choices considered in this work. Chapter 3 reviews previous work on Portuguese variety identification and rewriting, broader language-variety processing, sequential adaptation, and reward-based fine-tuning. Chapter 4 details the corpora, preprocessing procedures, filtering decisions, and stage-level data allocation. Chapter 5 describes the models, task formulations, training stages, adaptation strategies, and reward-based continuation setup. Chapter 6 presents the experimental methodology and analyzes the identification and rewriting results. Finally, Chapter 7 summarizes the main findings and outlines future research.

2

Fundamental Concepts

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This chapter introduces the linguistic and technical background needed to understand the modeling choices made in the remainder of this work. It first describes the main differences between European and Brazilian Portuguese (Section 2.1). It then introduces the modern language-model background needed later in this work, covering language modeling itself, the Transformer architecture, and the main model families amenable to fine-tuning (Section 2.2). The chapter then turns to the formulations used for Portuguese variety identification and rewriting (Section 2.3). It closes with the adaptation techniques used later in this work: Supervised Fine-Tuning (SFT) and Reinforcement Learning (RL)-based optimization, introduced in Sections 2.4 and 2.5.

2.1 Differences Between European Portuguese and Brazilian Portuguese

Before introducing the modeling background, it is useful to separate the linguistic phenomena that provide evidence for EP/BP identification from those that require edits during rewriting. These contrasts are not limited to isolated vocabulary substitutions. Prior literature on paraphrastic correspondences

between the two varieties shows that many differences appear as recurrent alternatives in how equivalent meanings are expressed, spanning lexical and morphosyntactic variation (Barreiro and Mota, 2018). Cyrino (2001) also discusses contrasts involving null objects and verb phrase ellipsis, relating them to changes in the verbal structure of BP sentences.

These observations are consistent with the quantitative analysis reported by Alves (2025), which studies EP/BP variation using information-theoretic measures. Lexical choice accounts for the largest divergence between the two varieties, but the research also identifies systematic grammatical preferences in subject-pronoun use, progressive constructions, and clitic-related patterns. Analyses based on part-of-speech distributions, dependency relations, and word-order entropy further suggest that EP and BP share most of their syntactic backbone, while still diverging in recurrent local preferences that are salient enough to affect automatic processing.

The following list highlights representative contrasts between the two Portuguese varieties that are relevant to identification and rewriting:

- **Orthography and morphology.** Orthographic differences can provide useful variety cues, especially in older or noisier corpora. However, many spelling contrasts were reduced by the Portuguese Orthographic Agreement, so forms such as *projecto*, *acto*, or *baptismo* are less reliable as markers in contemporary standard data (Costa-jussà et al., 2018).

- **Progressive and aspectual periphrases.** EP typically uses *estar a* + infinitive for the progressive (e.g., *Ele estava a correr*), whereas BP typically uses *estar* + gerund (e.g., *Ele estava correndo*) (Barreiro and Mota, 2018).

- **Clitic placement and word order.** EP commonly places clitics postverbally with hyphenation in affirmative contexts (e.g., *Ele viu-me na rua*), while BP more often favors preverbal placement without hyphenation (e.g., *Ele me viu na rua*) (Barreiro and Mota, 2018).

- **Subject pronoun realization (pronoun drop).** Both varieties allow null subjects, but BP uses overt subject pronouns more often than EP.
  For example, EP allows *Fui ao supermercado ontem*, whereas BP more often uses *Eu fui ao supermercado ontem* (Barreiro and Mota, 2018; Costa-jussà et al., 2018).

- **Pronoun preferences.** BP commonly uses *você* as a neutral or default second-person address form, whereas EP more often uses *tu*, nominal address forms, or null subjects depending on register. In rewriting, this can create frequent pairs between EP second-person verb morphology and BP third-person morphology with *você* (Barreiro and Mota, 2018; Costa-jussà et al., 2018).

- **Determiners with possessives and zero determiners.** In BP, definite articles before possessives are often omitted (e.g., *Vendi meu carro*),
  whereas EP more typically requires them (e.g., *Vendi o meu carro*). BP also permits zero determiners more often than EP (Barreiro and Mota, 2018; Cyrino, 2001).

- **Lexical and multiword preferences.** Some of the most robust distinctions in written corpora arise from lexical choice and multiword expressions, such as *comboio*/*trem*, *fato*/*terno*, or *toda a gente*/*todo o mundo* (Barreiro and Mota, 2018; Costa-jussà et al., 2018).

These phenomena affect the two tasks in different ways. For variety identification, lexical preferences and some orthographic forms often provide strong surface evidence. For rewriting, the structural phenomena are more important, requiring the system to decide whether to preserve the sentence, apply a local substitution, or make a constrained syntactic change while preserving meaning and scope.

2.2 Modern Language Models

This section introduces the modeling background needed for the remainder of the document. It begins with language modeling itself (Section 2.2.1), then describes the Transformer architecture and the distinction between encoders and decoders (Section 2.2.2). It then reviews the main Transformer architectures used for language tasks and explains why encoder-decoder models are an appropriate starting point for Portuguese variety identification and rewriting (Section 2.2.3).

2.2.1 Language Modeling

Language modeling is the task of assigning probabilities to sequences of text. Given a sequence of tokens $x_1, x_2, \ldots, x_n$, a language model estimates the probability of the full sequence by factorizing it into conditional probabilities, typically of the form $p(x_1, \ldots, x_n) = \prod_t p(x_t \mid x_{<t})$. In practice, this means the model learns which tokens are likely to occur in a given context, where $x_{<t}$ denotes the tokens that precede position t. In autoregressive settings, this factorization is implemented directly as next-token prediction. Given a prefix, the model predicts the most likely continuation one token at a time. In other self-supervised settings, the training signal is different, but the model still uses context to reconstruct portions of text and thereby learn regularities of the language (Zhao et al., 2026).

Early language models were predominantly statistical. A common example is the n-gram model, which estimates the probability of a token from the previous $n-1$ tokens by counting how often short sequences occur in a corpus. This approach is simple and effective for capturing local regularities, but it is limited by the fixed size of the context window. It also suffers from data sparsity, as many possible sequences occur rarely or not at all, even in large corpora (Rosenfeld, 2000; Jelinek, 1998).

Neural language models replaced explicit count-based statistics with learned distributed representations (Bengio et al., 2003). Instead of treating each context as an isolated symbolic pattern, these models learn dense token representations and trainable functions that can generalize across related contexts. Recurrent architectures, especially Long Short-Term Memory (LSTM) networks, extended this idea to variable-length sequences and made it possible to carry information across longer contexts, although efficient large-scale training remained difficult (Mikolov et al., 2010).

Language-modeling objectives can be applied without task-specific labels, which supports pretraining on large unlabeled corpora before adaptation to downstream tasks. Pretrained models therefore offer

a stronger starting point than training from scratch. The next subsection introduces the Transformer architecture, which underlies many state-of-the-art language models and is central to this study.

2.2.2 Transformers: Self-Attention, Encoders and Decoders

The central operation in the Transformer architecture is self-attention. Self-attention allows each token in a sequence to combine information from other positions by computing attention weights that reflect learned similarities between token representations. Each token is first mapped to a continuous embedding. Since self-attention is permutation-invariant, information about token order must be added separately through positional encodings. In the original Transformer, these are fixed sinusoidal vectors added to token embeddings, yielding input representations $X \in \mathbb{R}^{n \times d_{\text{model}}}$, where n is the sequence length and d_{model} is the hidden dimension. In practice, n is bounded by the model's context window, the maximum number of tokens it can attend to in one forward pass (Vaswani et al., 2017).

Given token representations X, which already include positional information, queries, keys, and values are obtained through learned linear projections,

$$Q = XW^Q, \qquad K = XW^K, \qquad V = XW^V. \tag{2.1}$$

The scaled dot-product attention used throughout the architecture is then given by

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V. \tag{2.2}$$

The operation replaces each token representation by a similarity-weighted combination of the others. Multi-head attention repeats this computation with different learned projections and concatenates the results, allowing the model to capture several interaction patterns in parallel (Vaswani et al., 2017).

Although many later architectures are based on self-attention, the original Transformer was proposed as an encoder-decoder model for sequence-to-sequence tasks. In the original architecture, the encoder consists of a stack of Transformer blocks. Each block contains a self-attention sublayer followed by a position-wise feed-forward network, where the same multilayer perceptron is applied independently to each token representation. Residual connections add the input of each sublayer to its output, allowing the network to refine previous representations instead of replacing them entirely. Layer normalization supports stable training. Since encoder self-attention is bidirectional, every position can attend to every other position in the source sequence. The result is a sequence of contextual representations that is useful both for text understanding and for conditioning downstream generation (Vaswani et al., 2017).

The decoder uses a similar stack, with two additional mechanisms required for generation. First, masked self-attention prevents each position from attending to future tokens, so generation proceeds from left to right. Second, cross-attention lets the decoder attend to encoder outputs, conditioning target generation on the source sequence (Vaswani et al., 2017).

From these building blocks, modern pretrained models organize encoders and decoders in different ways. The next subsection reviews the main families that result from these choices and explains why the encoder-decoder architecture is suitable for the tasks studied in this work.

2.2.3 Transformer Architectures for Language Tasks

Modern pretrained language models share the Transformer building blocks introduced above, but organize those blocks differently and pair them with different pretraining objectives. The main families relevant to the present work are encoder-only, decoder-only, and encoder-decoder models.

Encoder-only models such as Bidirectional Encoder Representations from Transformers (BERT) are bidirectional encoders trained to recover masked tokens or spans from surrounding context (Devlin et al., 2019). Since each position can attend to both left and right context, these models are especially effective at building contextual representations for tasks such as classification, retrieval, or span prediction. In practice, they are often adapted by attaching a task-specific prediction head on top of the encoder. Their limitation for the present work is that they are not generative by default, which makes them less convenient when one wants a single model family to cover both understanding- and generation-oriented tasks.

Decoder-only models use a single Transformer stack with causal masking and define an autoregressive distribution over continuations (Brown et al., 2020), which makes them a natural fit for open-ended generation. When they undergo additional adaptation through instruction tuning, they can also become well suited to task following (Chung et al., 2024). These models can be used for classification as well, either by generating a label directly or by scoring candidate labels according to their conditional likelihood.

Encoder-decoder models combine a bidirectional encoder with an autoregressive decoder. The encoder constructs contextual representations of the source sequence, and the decoder generates the target sequence while attending to that source representation. This architecture is especially suited to conditional generation, where the output must be produced from a given input. Machine translation (MT) is the classical example, with the model encoding a source sentence and decoding a target sentence. The same input-output structure can also be used for other text tasks. In variety identification, the output is a short label, whereas in rewriting, the output is a full sentence in the requested variety (Raffel et al., 2020).

All three families operate over tokenized text within a finite context window and can benefit from transfer learning through downstream fine-tuning. Modern systems often rely on subword units, which reduce out-of-vocabulary problems and help represent spelling or morphological variation (Sennrich et al., 2016). The relevant question is not whether these families can in principle represent the tasks studied, but how naturally their pretraining setup aligns with the desired use case. Encoder-only models are particularly strong for understanding-oriented prediction and decoder-only models provide a flexible generative interface. This work combines bidirectional rewriting and variety prediction, making encoder-decoder models a coherent starting point for the modeling choices that follow.

Among encoder-decoder options, T5Gemma 2 is a relevant model family for this project. It is built by taking pretrained decoder-only Gemma checkpoints and continuing pretraining in an encoder-decoder, text-to-text setting with a Unifying Language Learning Paradigms (UL2)-style objective (Zhang et al.,

2025b; Tay et al., 2023). This objective combines denoising tasks such as span corruption with prefix-style prediction, so the model uses bidirectional source context while retaining autoregressive generation.

For the tasks this project aims to study, the relevance of T5Gemma 2 is not limited to its general pretrained knowledge. It is also compatible with conditional generation and structured text targets, which are needed when a single framework must support both sentence rewriting and short label outputs.

Two architectural choices further motivate this selection. T5Gemma 2 shares word embeddings across the encoder and the decoder, and it replaces the usual separation between decoder self-attention and cross-attention with a merged attention module (Zhang et al., 2025b). These choices are intended to reduce duplication and improve efficiency while preserving access to the encoded source. Recent work on adapting Gemma models from decoder-only to encoder-decoder form further argues that this architecture offers a favorable trade-off between richer source representations, efficient inference, and strong downstream fine-tuning behavior (Zhang et al., 2025a).

2.3 Task Formulations for Portuguese Variety Identification and Rewriting

The standard formulation of variety identification is supervised classification where, given a text segment, the system selects one label from a small set of possible varieties, such as EP or BP (Zampieri and Gebre, 2012; Sousa et al., 2025b). In a generative model, the task can instead be represented as generation over a restricted output space in which the decoder produces a short variety label.

Variety rewriting is naturally formulated as conditional generation. The model receives a source sentence together with a target-variety specification and generates an equivalent sentence in the requested variety. The output should preserve meaning while making the changes required by the target variety.

A shared sequence-to-sequence interface can represent both tasks within the same model (Raffel et al., 2020). The two tasks rely on overlapping variety-sensitive evidence: cues that support an EP/BP decision are often also cues that must be preserved or modified during rewriting. Training on both tasks can therefore allow identification data to reinforce variety-sensitive signals while rewriting data teaches how those signals should change when a target variety is requested, without implying that a multitask model must outperform a dedicated single-task model on every metric (Caruana, 1997).

In the formulations studied in this work, rewriting direction is represented through an input-variety label. One option is to place this information in the input, so the encoder receives it together with the source sentence. Another option is to place it in the output structure, so the decoder is required to produce the variety label before the rewritten sentence. These alternatives leave the underlying tasks unchanged, but change where the control signal is introduced. The exact labels, token choices, and dataset conversion rules are part of the experimental design and are described in Chapter 5.

2.4 Fine-Tuning Pretrained Language Models

Pretrained models provide a strong starting point, but they do not necessarily exhibit the behavior required by a downstream task. Even with broad linguistic knowledge, a model may not consistently distinguish varieties, preserve meaning during rewriting, or follow the requested target variety. Fine-tuning addresses this gap by continuing training on data that is closer to the intended use case.

In SFT, the model is trained on input-output examples and its parameters are updated to increase the likelihood of the desired targets. For a text-to-text model, both classification and rewriting examples can be represented as sequence prediction problems (Raffel et al., 2020).

SFT does not have to be carried out in a single phase. When different sources of supervision play different roles, a model can first be adapted on broader data and then continued on more targeted data. A staged setup is common in Neural Machine Translation (NMT) domain adaptation, where broad parallel data is often followed by in-domain fine-tuning (Cao et al., 2021; Gu et al., 2022). For the present work, the same idea motivates separating broader rewriting adaptation, more targeted EP/BP supervision, and later reward-based optimization. The main risk is that later phases can pull the model toward newly introduced data and weaken behavior learned earlier (Gu et al., 2022; Gu and Feng, 2020).

Another central design choice is how many parameters should be updated during adaptation. Full fine-tuning updates the entire pretrained model. Updating all weights gives the optimizer maximum flexibility and can be useful when the target task or domain differs substantially from pretraining, but it is expensive in memory and computation, and may overfit when the supervision is narrow. Parameter-Efficient Fine-Tuning (PEFT) instead freezes most or all of the pretrained base model and learns a much smaller set of task-specific parameters. Adapter-based methods insert small trainable modules into the network while keeping the original model fixed, allowing task adaptation with far fewer updated parameters than full fine-tuning (Houlsby et al., 2019; Pfeiffer et al., 2021).

LoRA is a widely used PEFT method. It implements parameter efficiency by learning low-rank updates to selected weight matrices rather than by inserting separate adapter layers. If a pretrained weight matrix is denoted by $W \in \mathbb{R}^{d\times k}$, LoRA keeps W frozen and learns an update $\Delta W = BA$, where $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, and r is much smaller than d and k. The effective adapted weight is $W + \Delta W$, but only the low-rank factors are trained (Hu et al., 2022). In the common LoRA parameterization, the new branch is scaled by α/r before being added to the frozen base projection. The rank r controls the capacity of the learned update. Smaller ranks impose a stronger bottleneck, whereas larger ranks allow a more expressive adaptation. The scaling factor α controls the magnitude of the LoRA contribution, and dropout can be applied to the adapter branch as regularization.

Another LoRA design choice is which Transformer projections receive low-rank updates. Early LoRA work often focused on attention projections for parameter efficiency (Hu et al., 2022), while later work showed that broader coverage can be useful when stronger task adaptation is required. In particular, Quantized Low-Rank Adaptation (QLoRA) reports that adapting all Transformer linear layers is important for recovering performance comparable to strong full-fine-tuning baselines (Dettmers et al., 2023). Broader coverage allows the adapters to modify both attention interactions and feed-forward transformations while

the pretrained weights remain frozen. This distinction is relevant to the experimental setup in Chapter 5, where LoRA is applied to both attention and feed-forward projections.

2.5 Reinforcement Learning and Reward-Based Fine-Tuning

Supervised fine-tuning optimizes a model to reproduce reference outputs token by token. Token-level supervision is useful when a reference target is available, but some later training objectives are easier to express as preferences or scalar scores over complete outputs. A generated sentence may be fluent while failing to follow an instruction, preserve meaning, respect a target variety, or avoid an undesirable behavior. Reward-based fine-tuning addresses this limitation by assigning a score to one or more candidate outputs and updating the model so that higher-scoring outputs become more likely (Ouyang et al., 2022).

A useful abstraction for this setting is the contextual bandit view (Sutton and Barto, 2018; Lattimore and Szepesvári, 2020). Classical multi-armed bandits model repeated decisions between actions whose reward distributions are initially unknown. After each decision, the agent observes only the reward of the action it selected. There is no long-term planning and no state transition caused by the action. Contextual bandits extend this setting by conditioning the decision on side information. At each round, a context X is sampled from a distribution $\mathcal{D}$, the agent chooses an action $a \sim \pi(\cdot \mid X)$, and observes a scalar reward $r(X, a)$. The objective is to learn a policy π that maximizes expected reward,

$$J(\pi) = \mathbb{E}_{X \sim \mathcal{D}} \mathbb{E}_{a \sim \pi(\cdot|X)} \left[r(X, a) \right]. \tag{2.3}$$

In reward-based fine-tuning for language models, the context X corresponds to the conditioning information given to the model, such as an instruction and a source sentence. The action is not a single symbolic choice, but a complete generated output $Y = (Y_1, \ldots, Y_{|Y|})$. A language model parameterized by θ defines the following autoregressive policy over complete outputs:

$$\pi_\theta(Y \mid X) = \prod_{t=1}^{|Y|} \pi_\theta(Y_t \mid X, Y_{<t}). \tag{2.4}$$

After sampling $Y \sim \pi_\theta(\cdot \mid X)$, the output is evaluated by a scoring function, such as a reward model, verifier, preference scorer, automatic metric, or task-specific evaluator. The score is an outcome-level reward $r(X, Y)$, and the contextual-bandit objective for language-model policy optimization becomes

$$J(\pi_\theta) = \mathbb{E}_{X \sim \mathcal{D}} \mathbb{E}_{Y \sim \pi_\theta(\cdot|X)} \left[r(X, Y) \right]. \tag{2.5}$$

The objective above can be optimized from different forms of feedback. Pairwise-preference methods such as Direct Preference Optimization (DPO) use preferred and rejected outputs for the same input and increase the relative probability of the preferred output (Rafailov et al., 2023). In sampled-output optimization, the model instead generates candidate outputs, assigns them scalar rewards, and updates the policy according to the rewards obtained by those candidates.

Within sampled-output optimization, group-relative methods compare several candidates generated for the same input and convert their rewards into relative advantages (Shao et al., 2024). If a group contains C candidates with rewards $r_1, \ldots, r_C$, a standardized group-relative advantage can be written as

$$\hat{A}_i = \frac{r_i - \bar{r}}{\sigma_r + \epsilon}, \qquad \bar{r} = \frac{1}{C}\sum_{j=1}^{C} r_j, \qquad \sigma_r = \sqrt{\frac{1}{C}\sum_{j=1}^{C}(r_j - \bar{r})^2}. \tag{2.6}$$

A candidate receives a positive advantage when it scores above the group average and a negative advantage when it scores below it. The optimization signal therefore depends on how a candidate performs relative to alternatives for the same input rather than on the absolute scale of the reward.

Once generated candidates can be scored, the next design question is what the reward should measure. In language-variety rewriting, an output may need to be fluent, faithful to the source, stylistically appropriate, and aligned with a requested constraint. Recent work on language-model alignment studies how multiple reward models can be transformed and combined when optimizing for several properties at once (Wang et al., 2024). A common approach combines reward components into a scalar objective,

$$R(X, Y) = \sum_{m=1}^{M} \lambda_m r_m(X, Y), \tag{2.7}$$

where r_m are component rewards and λ_m are weights controlling their relative importance. Reward shaping can also transform or augment these components, for example by normalizing them, comparing them to a baseline, or adding penalties for undesirable behaviors.

3

Related Work

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This chapter reviews previous work that is most directly connected to the tasks studied in this research. Section 3.1 covers EP/BP identification, while Section 3.2 reviews work on rewriting between the two varieties. Section 3.3 broadens the discussion to language processing beyond Portuguese, including work on dialectal and regional settings. Section 3.4 reviews staged adaptation for machine translation, and Section 3.5 covers reward-based adaptation for translation. Section 3.6 identifies the research gap.

3.1 European and Brazilian Portuguese Identification

Early work on EP/BP identification showed that simple surface features are already highly informative for this distinction. Zampieri and Gebre (2012) proposed one of the first automatic systems for discriminating European and Brazilian Portuguese, using character n-grams and word n-gram language models. Their results indicate that orthographic and lexical cues can be highly informative for EP/BP identification. More broadly, work on document language identification cautions against treating closely related varieties as an easy identification problem (Da Silva and Lopes, 2006).

Castro et al. (2016) tested the same distinction on Twitter, where spelling variation, limited context, and informal writing make EP/BP identification less controlled than in edited corpora. Their use of smoothed n-gram models and an ensemble strategy shows that lexical and character-level evidence remains useful in social-media settings, although its reliability depends on the amount and quality of the available context. Preda et al. (2024) introduce stronger experimental controls through a Portuguese dialect-identification dataset derived from TED2020, a parallel corpus containing aligned European and Brazilian Portuguese versions of the same talks. The authors compare handcrafted linguistic features, word n-gram classifiers with optional part-of-speech (POS) information, adaptive Naive Bayes variants, and LoRA-adapted models. Their evaluation varies input length, from sentence-level segments to full transcripts, and tests cross-dataset robustness on out-of-domain news sources, which makes text granularity and domain shift part of the evaluation rather than minor experimental settings. The LoRA-adapted Albertina classifiers obtain the strongest in-domain results on TED2020, while a logistic-regression bigram model performs best in the joint cross-dataset evaluation.

The reliability of these comparisons depends heavily on dataset construction, particularly when labels are inferred from metadata such as source country, domain, or Uniform Resource Locator (URL) (Zampieri et al., 2014). For very similar varieties, metadata can be weak supervision, as a text collected from a given country may contain few explicit markers of that variety, or may have been written in a different variety altogether. Zampieri et al. (2024) address this issue with Discriminating Between Similar Languages – True Labels (DSL-TL), a human-annotated multilingual dataset for language-variety identification that includes European and Brazilian Portuguese, highlighting that high scores on silver-labeled corpora do not necessarily imply robust variety discrimination.

More recently, Sousa et al. (2025b) compiled PtBrVarId, a cross-domain dataset covering journalistic, legal, political, web, social-media, and literary texts, and evaluated both n-gram-based baselines and Transformer-based classifiers. Their best systems fine-tune BERTimbau (Souza et al., 2020) and use a two-step cross-domain protocol with partial delexicalization, masking named entities and selected part-of-speech categories to reduce overfitting to domain-specific lexical cues. The delexicalized BERT model is their strongest system on both DSL-TL and FRMT. Their FRMT identification results provide one of the closest published comparisons for the identification task in this work.

3.2 Rewriting Between European and Brazilian Portuguese

Marujo et al. (2011) studied the specific task of converting BP texts into EP, rather than generating Portuguese text more generally. Their rule-based and statistical setting was used to adapt BP resources for EP-oriented machine translation, and the adapted resources improved translation quality compared with using the same material without the BP-to-EP conversion step. A complementary linguistic perspective is provided by Barreiro and Mota (2018), who extracted paraphrastic correspondences between both Portuguese variants from a parallel corpus. Their analysis shows that some EP/BP correspondences involve multiword expressions and syntactic changes, so rewriting is not always word replacement.

In the neural setting, Costa-jussà et al. (2018) presented a system for translating between EP and BP in both directions and compared it with phrase-based statistical machine translation. Their recurrent encoder-decoder model improved automatic scores and was also preferred in human evaluation, indicating that neural models can learn cross-variety transformations that challenge phrase-based methods.

Sanches et al. (2024) bring this task closer to current pretrained systems by focusing on BP-to-EP translation with modern encoder-decoder and decoder-only architectures. The authors provide a comparison point for the rewriting component of this work by fine-tuning multilingual encoder-decoder models, such as M2M100 and mBART50, on parallel data from subtitles and TED talks. They introduce the manually curated Golden Collection benchmark and compare the fine-tuned models with ChatGPT 3.5 using both automatic metrics and human evaluation. Their study finds that the best fine-tuned system generalizes well on test data close to its training domains, while ChatGPT 3.5 is stronger on the manually curated Golden Collection. Copying the input is also a very strong baseline against the manual references in that collection, reflecting the conservative nature of many valid EP/BP conversions.

3.3 Language Processing Beyond Portuguese

Language- and variety-identification systems can affect not only evaluation but also the data available for later model training (Zampieri et al., 2020). OpenLID (Burchell et al., 2023) and OpenLID-v3 (Fedorova et al., 2026) provide multilingual language-identification resources for data curation, where predictions can determine whether candidate web texts are kept, discarded, or routed to a language-specific subset. FastSpell (Bañón et al., 2024) specifically addresses confusable languages by combining an initial language-identifier prediction with dictionary-based verification. Caswell et al. (2020) show why this distinction matters in practice: strong benchmark F1 does not necessarily produce clean web-filtered corpora under domain mismatch, class imbalance, and language similarity. Identification errors can therefore propagate into the composition of the corpora used for subsequent adaptation.

Related work on generation shows that national and regional varieties can also be represented explicitly as output targets rather than collapsed under a single language label. Lakew et al. (2018) train multilingual NMT systems from English into specific national varieties and closely related standardized languages, including European and Brazilian Portuguese, and find that treating the varieties as distinct targets improves translation consistency in their setting. The task remains translation from English rather than rewriting directly between two Portuguese varieties. Sousa et al. (2025a) provide a more recent Portuguese example with PTradutor, an open-source English-to-EP translation model built with retro-translated data. Its improvement over generic open-source Portuguese translation systems further shows that targeting the desired national variety can matter during data construction and model adaptation.

Broader dialect and language-variety benchmarks show that the usefulness of different model families depends on the task being evaluated. Faisal et al. (2024) introduce DIALECTBENCH, a large-scale benchmark covering 281 varieties and 10 text-level tasks. Building on this benchmark, Faisal and Anastasopoulos (2025) compare instruction-tuned LLMs with supervised encoder-based models and find that LLMs can be competitive on some comprehension-style tasks, while multi-class identification often

still benefits from task-specific supervised fine-tuning. The two studies therefore provide a benchmark and a subsequent model comparison within the same broader evaluation setting, rather than independent evidence about unrelated dialect-processing problems.

Work that combines identification with generation is particularly relevant to the shared formulation studied in this work. Khered et al. (2025) combine Arabic dialect identification and dialect-to-Modern-Standard-Arabic translation within a multi-task encoder-decoder model. Their experiments report improved translation over the evaluated baselines, but the study changes both the shared multi-task setup and the training data, so the gain supports the combined experimental design without isolating multi-task learning itself as the cause. Sharma et al. (2026) introduce INDIC-DIALECT, a human-curated corpus and multi-task benchmark for Hindi and Odia dialects that includes dialect classification, multiple-choice question answering, and machine translation. Their experiments find weak zero-shot LLM classification, substantially stronger fine-tuned Indic models, and translation results in which hybrid rule-based and neural systems perform best, with the preferred strategy depending on conversion direction. Together with the preceding work, these studies place the present EP/BP setting within a broader literature in which variety information affects corpus selection, generation targets, and joint task modeling.

3.4 Staged Adaptation for Machine Translation

Classical NMT domain adaptation trains models on general-domain bitext before adapting them to in-domain data, assigning broad and specialized supervision different roles across successive phases. Xu et al. (2024a) introduce the Advanced Language Model-based trAnslator (ALMA), a decoder-only MT system adapted first with monolingual data and then with a smaller set of high-quality parallel data. Their ablations show benefits from the monolingual adaptation stage and from high-quality parallel supervision, without isolating the effect of stage order.

Alves et al. (2024) use a related staged recipe in Tower, combining continued pretraining on multilingual monolingual and parallel data with later fine-tuning on translation-related instructions. Rei et al. (2026) extend this recipe by adding preference optimization and reinforcement learning under verifiable reward signals, so later phases introduce objectives that are absent from the earlier supervised stages. Omnilingual MT (Omnilingual MT Team et al., 2026) assigns distinct roles to several training phases at larger scale. Its decoder-only system starts from LLaMA instruction models, extends the tokenizer for multilingual coverage, continues pretraining on monolingual language modeling and parallel translation data, and then uses supervised fine-tuning to recover instruction-following behavior while specializing the model for MT. Its encoder-decoder OMT-NLLB system uses decoder training on parallel MT and monolingual autoencoding, cross-attention warm-up, and final end-to-end fine-tuning on parallel data. The reported gains from autoencoding, warm-up, and the final stage show that successive phases can serve different functions rather than simply extending the total amount of training.

Sequential specialization can also weaken behavior acquired in earlier training phases. Continual-learning studies examine this retention problem explicitly in NMT (Cao et al., 2021; Gu and Feng, 2020; Gu et al., 2022). Shao and Feng (2022) show that related degradation can also arise from

imbalanced exposure to training examples outside a strict continual-learning setting. These results make retention across later adaptation phases an important consideration without treating staged adaptation and continual learning as the same setting.

3.5 Reward-Based Adaptation for Translation

Nguyen et al. (2017) show that an existing NMT system can be updated from scalar feedback without receiving a corrected translation for every generated sentence. Their bandit NMT experiments use simulated human ratings and optimize corpus-level MT objectives with policy-gradient updates. Thu et al. (2023) apply reward-based continuation after supervised training for low-resource Burmese dialect translation and report improvements over non-RL baselines. In both studies, reward optimization modifies an already trained translation system rather than defining the translation task from scratch.

He et al. (2024) use Quality Estimation (QE), which predicts translation quality without requiring a reference translation, as a reward for MT feedback training. They report translation-quality gains but also observe overoptimization, where the learned reward continues to improve while actual translation quality deteriorates. Xu et al. (2024b) train a reward model from human translation preferences and report improvements over their supervised starting model after Reinforcement Learning from Human Feedback (RLHF)-based translation optimization, together with evidence that part of the learned preference signal transfers to translation directions not used for RLHF training.

Ramos et al. (2026) use xCOMET error spans and severity predictions to replace a single sentence-level score with token-level rewards. Their experiments report better translation quality than sentence-level reward baselines and more stable reward progression during training. Feng et al. (2025) combine format constraints with metric-based translation quality in a mixed reward and show that lexical, semantic, and mixed metric rewards lead to different translation behavior during continuation. Wang et al. (2024) show that reward scaling, transformations, and aggregation affect behavior under joint optimization.

Omnilingual MT (Omnilingual MT Team et al., 2026) reports an optimization failure in which straightforward Group Relative Policy Optimization (GRPO) continuation after SFT produces low entropy and vanishing gradients. The revised setup enlarges candidate groups, reintroduces Kullback–Leibler (KL) regularization, and uses a balanced multi-criterion reward. Liu et al. (2026) show a complementary failure mode in multilingual MT, where source-based QE rewards can amplify weaknesses in the evaluator. These results show that reward-based MT can fail through either reward design or optimization.

3.6 Overview and Research Positioning

The closest published experimental comparisons are Sousa et al. (2025b) for EP/BP identification on FRMT and Sanches et al. (2024) for BP-to-EP rewriting on the Golden Collection. Neither study evaluates identification and bidirectional rewriting across both resources. Sousa et al. (2025b) provide the most direct published identification comparison used in Chapter 6, while Sanches et al. (2024) provide the most direct published rewriting comparison and introduce the Golden Collection benchmark.

In Arabic, Khered et al. (2025) combine dialect identification with dialect-to-Modern-Standard-Arabic translation in a shared encoder-decoder model. Their translation gains are reported for a setup that also changes the available training data, so they do not isolate the contribution of multi-task learning alone. Khered et al. (2025) study one-way dialect-to-standard translation, whereas this work combines variety identification with bidirectional EP/BP rewriting and alternates between short label outputs and full-sentence generation in the same model.

Caswell et al. (2020) show that language-identification scores can overstate corpus quality when labels are noisy, classes are imbalanced, or closely related varieties are mixed. Staged MT studies show that successive adaptation phases can serve different roles and that later training can alter earlier behavior (Alves et al., 2024; Rei et al., 2026; Omnilingual MT Team et al., 2026). Reward-based MT studies show that evaluator choice, reward composition, and optimization stability can materially affect the behavior learned during continuation (He et al., 2024; Feng et al., 2025; Wang et al., 2024).

Within the literature reviewed in this chapter, no single study combines EP/BP identification and bidirectional rewriting in a shared model while evaluating both tasks across FRMT and the Golden Collection. This work studies that combined setting with encoder-decoder models and examines how encoder-side and decoder-side control, supervision composition, staged adaptation, and reward-based continuation affect identification and conservative rewriting. The rewriting objective is to preserve meaning while making only the changes needed for the requested Portuguese variety.

4

Data

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The reported experiments use four main resources with different roles. Each resource is first converted into a common aligned-pair format containing a BP sentence, an EP sentence, optional dataset-specific metadata, and a split label. This chapter describes the origin and preparation of each resource. Table 4.1 summarizes aligned EP/BP pair counts for each resource before task-specific rendering.

Table 4.1: Aligned EP/BP pair counts by resource and split before task-specific rendering.

Resource	Train	Validation	Test
OpenSubtitles	10,347,883	0	0
GPT-Wikipedia	4,519	564	566
FRMT	2,502	20	2,611
Golden Collection	0	0	500

4.1 OpenSubtitles

OPUS OpenSubtitles (Tiedemann, 2009) is a large collection of aligned subtitles from movies and television series. It provides the scale needed for an initial broad adaptation phase and exposes the models to diverse topics and informal dialogue. However, its scale comes with limitations, as subtitle segments can be elliptical, weakly aligned, or stylistically distant from conservative EP/BP adaptation. Its role in the experiments is broad exposure rather than precise control of individual variety markers.

After exact duplicate pairs are removed, preprocessing first removes subtitle-specific formatting noise, including leading dialogue dashes, speaker labels, dash artifacts after punctuation, irregular whitespace, wrapping and unbalanced quotation marks, and fully capitalized line formatting. The next step repairs segmentation when adjacent subtitle rows appear to belong to the same sentence. Rows i and $i+1$ are merged when either the EP or BP side provides a continuation cue: row i ends in a comma, semicolon, colon, or dash, or contains fewer than 40 characters, and the first alphabetic character of row $i+1$ is lowercase. A separate merge condition handles cases where the paired sentences show inconsistent apparent sentence boundaries. Whenever a merge condition is met on either side, both BP and EP rows are joined so their alignment is preserved. Table 4.2 illustrates representative segmentation repairs. In each example, "Line i" and "Line $i+1$" are adjacent subtitle rows, while "Merged sentence" shows the resulting BP text.

Table 4.2: Examples of OpenSubtitles segmentation repairs using the BP side of each aligned EP/BP pair.

Merge cue	Line i	Line $i+1$	Merged sentence
Unfinished punctuation + lowercase continuation	*Não achava que o basquete era assim, o cansaço, a vitória,*	*o cansaço.*	*Não achava que o basquete era assim, o cansaço, a vitória, o cansaço.*
Short fragment split across rows	*Massa Macinaia.*	*Deve ser uma vila perto de...*	*Massa Macinaia. Deve ser uma vila perto de...*
Ellipsis + lowercase continuation	*Não pode colocar na sua cabeça? ...*	*eu não morri na primeira vez.*	*Não pode colocar na sua cabeça?... Eu não morri na primeira vez.*

Surface overlap alone is not a reliable filter for aligned EP/BP pairs, since valid correspondences may preserve the same content while using different lexical forms or one-to-many constructions. For example, a BP gerund can correspond to an EP infinitival construction. The pairs are therefore filtered with SimAlign (Jalili Sabet et al., 2020), which uses contextual token embeddings to provide a bilingual alignment signal rather than relying on exact word matching.

Using SimAlign requires choosing how the alignments are obtained. The first choice is the multilingual encoder used to compute contextual embeddings for both sentences. The second is the granularity of the alignment, since token connections can be produced over subword pieces or over the original words after combining the representations of their subword pieces. The third is the alignment extraction method that converts token similarities into links between the sentences of the pair. In this work, the filter uses the XLM-RoBERTa (XLM-R)-based encoder configuration, word-level alignments, and the `inter` extraction method. This extraction can be understood as a similarity matrix whose rows are BP tokens and whose columns are EP tokens. Each cell estimates how close two contextual token representations are, and `inter` keeps a link only when the two tokens select each other as their highest-scoring option. The resulting links are used only as evidence that the paired sentences still express the same content. They are converted into a coverage score over normalized word tokens. Shorter subtitle fragments are allowed to have lower coverage, while longer pairs must show stronger bilingual alignment. A separate length-ratio check rejects pairs where one sentence is much shorter or longer than its counterpart. The exact scoring rule, thresholds, alternative SimAlign options, and an illustrative token-similarity matrix are detailed in Appendix A.

4.2 GPT-Wikipedia

GPT-Wikipedia is the main constructed resource for controlled EP/BP rewriting. It was created to provide more conservative supervision than could be expected from subtitle data alone, while retaining broader topical coverage than a small manually curated benchmark. The resource is designed around near-literal EP/BP pairs, where the two sentences remain close in meaning and word overlap. Stricter conservative subsets are also derived from this resource for later reward-based continuation experiments.

GPT-Wikipedia is built from Portuguese Wikipedia article dumps used as inspiration material. For each article, only the first few sentences from informative paragraphs are kept. Paragraphs that are too short, list-like, or punctuation-poor are discarded. These texts are used only to provide topics and contexts for generation, not as aligned examples.

Generated pairs are obtained with a Portuguese prompt sent to OpenAI GPT-4o through the IAedu API. The template requests new EP/BP pairs in JSON format and favors minimal lexical change. Its main constraints are summarized by the translated and simplified excerpt below.

> Generate N new bilingual examples in valid JSON, with one EP sentence and one BP sentence per example. Use the reference passages only as inspiration for topic, domain, and register. Keep the two variants as close as possible. Do not write free paraphrases or change meaning without need. Change only clear lexical or morphosyntactic contrasts. In non-identical pairs, prefer one or two clear differences and avoid extensive rewrites.

The generated batches are then merged and checked for suspicious differences. The check compares observed edits with an allowlist of expected EP/BP correspondences and flags unexpected lexical substitutions or broader reordering for review.

During review, variety-relevant differences are kept with the EP expression in the EP sentence and the BP expression in the BP sentence. Differences judged unnecessary for EP/BP adaptation are collapsed to the same wording. As a result, the final cleaned resource is not restricted to non-equal pairs. The dataset contains 1,648 equal pairs among 5,649 pairs after cleanup.

4.3 FRMT

FRMT (Riley et al., 2023) is a benchmark for few-shot region-aware MT. It provides English sentences translated into regional varieties, including EP and BP. Its examples are organized into three buckets. The lexical bucket is designed around articles likely to elicit region-dependent terminology. The entity bucket focuses on people, places, institutions, and other region-associated entities. The random bucket draws from randomly selected high-quality Wikipedia articles and evaluates broader regional behavior without preselecting a lexical or entity contrast.

FRMT is not generated internally, but it is converted into the same pair format used by the rest of the resources so that training and evaluation scripts can share a common interface. Its original bucket labels are retained as metadata, although the reported evaluation aggregates over the complete FRMT split.

The complete FRMT training split provides additional supervised data in the GPT-Wikipedia + FRMT condition, while the held-out test split is used for evaluation. A stricter conservative subset is selected later for reward-based continuation, as described in Section 5.3.3.

4.4 Golden Collection

The Golden Collection is a manually curated evaluation resource introduced by Sanches et al. (2024). It contains 500 BP sentences divided across Science, Legislative, Literature, Social Media, and Wiki topics. In the original dataset, each source is associated with EP references, including a manually curated reference and an automatic DeepL reference.

Each BP sentence is paired with the manually curated EP reference, rather than the automatic DeepL reference. The resulting 500 EP/BP pairs are used only for evaluation. For rewriting evaluation, each pair is converted into two examples: one BP-to-EP and one EP-to-BP. When both sentences are identical, the expected output is the unchanged sentence.

5

Methodology

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This chapter describes the experimental methodology. Section 5.1 introduces the model family and the fine-tuning regimes. Section 5.2 explains how rewriting and identification instances are represented in a shared sequence-to-sequence format. Section 5.3 presents the three-stage fine-tuning pipeline. Section 5.4 defines the supervised protocol-selection study preceding the 4B-parameter experiments.

5.1 Models and Fine-Tuning Setup

The reported experiments use encoder-decoder models from the T5Gemma 2 family (Zhang et al., 2025b) with 270M and 4B parameters. The 270M-parameter experiments start from `google/t5gemma-2-270m-270m`, and the 4B-parameter experiments start from `google/t5gemma-2-4b-4b`.

The 270M-parameter model is fully fine-tuned for the supervised protocol-selection study described in Section 5.4. These lower-cost experiments compare choices involving batch composition, control-string treatment, and classification loss before the experiments with the 4B-parameter model are conducted. The study is not a controlled comparison between full fine-tuning and LoRA, since model scale and adaptation method change at the same time.

The 4B-parameter model is adapted with LoRA using rank $r = 48$, scaling factor $\alpha = 96$, and dropout $= 0.05$. LoRA is applied to the attention and feed-forward projection layers, while bias terms remain

frozen. With $r = 48$, this setup yields 187,772,928 trainable LoRA parameters. A separate LoRA-rank diagnostic is reported in Appendix B.

The reported supervised experiments use seed 42. During supervised training, encoder inputs and decoder targets are truncated to a maximum of 512 tokens during preprocessing.

5.2 Task Formulations and Instance Construction

Two control formulations are used to construct rewriting and identification instances: encoder-controlled and decoder-controlled. Both use PT and BR, while CLS appears only in encoder-controlled identification inputs. PT marks an EP source sentence, BR marks a BP source sentence, and PT and BR determine rewriting direction and act as identification targets.

An aligned EP/BP pair can provide supervision for both tasks. For rewriting, each sentence in the aligned pair is used once as input and once as output. One aligned pair therefore produces a BP-to-EP instance and an EP-to-BP instance. For identification, each sentence is treated independently and paired with its variety label. A complete non-equal pair can consequently provide one BP identification instance and one EP identification instance. Equal pairs, that is, aligned EP/BP pairs whose two sentences are identical, are retained for rewriting, but they do not generate identification rows. Such rows would assign different variety labels to identical sentence text. The templates used to build these examples are summarized in Table 5.1. These templates correspond to the task-specific row types used during training. Section 5.3.1 and Section 5.3.2 quantify the rewriting and classification rows in each stage mixture.

In both formulations, rewriting and identification share the same encoder-decoder parameters, embeddings, and decoder vocabulary. The formulations differ in control-signal placement and in how classification supervision is attached to the target sequence.

5.2.1 Encoder-Controlled Formulation

For BP-to-EP rewriting, the encoder receives the BP source sentence prefixed with BR, and the decoder target is the aligned EP sentence. For EP-to-BP rewriting, the encoder receives the EP source sentence prefixed with PT, and the decoder target is the aligned BP sentence. In the encoder-controlled formulation, identification uses the input CLS followed by the sentence, and the decoder target is the corresponding variety label. In this task, the complete label target is supervised.

A matched rewriting-only configuration is defined as an ablation of this standard encoder-controlled formulation. It uses the same rewriting templates but omits identification examples during training.

5.2.2 Decoder-Controlled Formulation

The alternative formulation moves the control signal to the generated sequence. In rewriting instances, the encoder receives the clean source sentence with no task prefix. For BP-to-EP rewriting, the decoder target starts with BR and then continues with the EP rewrite. For EP-to-BP rewriting, the decoder target

starts with PT and then continues with the BP rewrite. Identification examples use the same decoder-side labels but omit the rewritten sentence, leaving PT or BR as the complete target.

The coupling is particularly direct in the decoder-controlled formulation. Both task types supervise the same first decoding decision, while rewriting additionally requires generation of the converted sentence.

The decoder-controlled setup also requires a distinction in the loss mask. Identification rows compute loss only for the first token of the tokenized decoder target. PT and BR are already present in the tokenizer's base vocabulary and each is represented as a single token in the tested configurations, so this position corresponds to the complete variety label. Rewriting rows retain the complete target sequence in the loss. This masking isolates classification from sequence termination and avoids teaching the decoder to treat the control string alone as a complete rewriting output.

Table 5.1: Sequence-to-sequence templates used to construct rewriting and identification instances. PT and BR denote the input variety, while CLS marks identification inputs.

Setup	Encoder input	Decoder target
Encoder-controlled rewriting	BP-to-EP: BR + BP sentence EP-to-BP: PT + EP sentence	BP-to-EP: EP sentence EP-to-BP: BP sentence
Encoder-controlled identification	CLS + sentence	PT or BR
Decoder-controlled rewriting	BP-to-EP: BP sentence EP-to-BP: EP sentence	BP-to-EP: BR + EP sentence EP-to-BP: PT + BP sentence
Decoder-controlled identification	sentence	PT or BR

5.3 Training Stages

The training procedure is organized into three stages. Stage A provides initial supervised adaptation, Stage B continues supervised training with targeted resources, and Stage C optionally applies reward-based fine-tuning to the Stage B models. The supervised protocol used in Stages A and B was selected beforehand with the 270M-parameter model in the study described in Section 5.4. Alongside this staged procedure, matched variants are used to isolate selected training choices and are introduced in the stage where they first apply. Both the 270M- and 4B-parameter models use Stages A and B, whereas the Stage C configurations use the 4B-parameter model. Separate experiments with the 270M-parameter model, reported in Appendix C, probe isolated reward and candidate-group design choices. These experiments serve as diagnostics rather than a selection procedure for the 4B-parameter Stage C configuration.

Across the supervised stages, optimization uses AdamW (Loshchilov and Hutter, 2019) with weight decay 0.01. The learning rate follows a linear schedule with warm-up, rising to the stage-specific peak value and then decreasing over the remaining training steps. The subsections give learning rates, warm-up budgets, gradient accumulation, training-step budgets, data composition, and validation procedures.

Table 5.2 summarizes the task-specific examples available for training at each stage. Stages A and B combine rewriting and identification supervision, whereas Stage C uses rewriting-only training data.

Table 5.2: Task-specific training examples by stage and data condition, with task shares shown in parentheses.

Stage	Data	Rewriting examples	Identification examples
A	OpenSubtitles	20,695,766 (73.0%)	7,666,883 (27.0%)
B	GPT-Wikipedia	9,038 (57.6%)	6,652 (42.4%)
B	GPT-Wikipedia + FRMT	14,042 (54.7%)	11,616 (45.3%)
C	GPT-Wikipedia	6,686 (100%)	—
C	GPT-Wikipedia + FRMT	6,828 (100%)	—

5.3.1 Stage A: Broad Variety Adaptation

Stage A provides broad supervised adaptation to both Portuguese varieties using OpenSubtitles as its only data source. OpenSubtitles provides substantially larger-scale EP/BP supervision than the resources used in Stage B. The resulting checkpoint starts the more targeted Stage B adaptation.

After task construction, Stage A follows the natural OpenSubtitles task mixture summarized in Table 5.2, with approximately 73% rewriting and 27% identification supervision. Under the supervised protocol selected in Section 5.4, the two tasks are not explicitly rebalanced.

The 4B-parameter model uses batch size 1, gradient accumulation 32, a peak learning rate of 10^{-5}, 400 warm-up steps, and a maximum of 2500 training steps. The protocol-selection experiments with the 270M-parameter model introduced in Section 5.1 use batch size 1, gradient accumulation 16, a peak learning rate of 2×10^{-5}, 200 warm-up steps, and a fixed budget of 2500 training steps. If the complete training budget is reached, these settings correspond to at most 80,000 training-example presentations for the 4B-parameter model and 40,000 for the 270M-parameter model. Under this protocol, training examples are shuffled in a seed-controlled order. These budgets cover only a small fraction of the Stage A training pool, so examples do not repeat before the maximum step budget is reached.

OpenSubtitles has no resource-level validation split. Stage A therefore uses a fixed 200-instance validation view sampled with seed 42 from the processed training data after task construction. The sampled instances remain eligible for training, so the validation view is not a disjoint held-out split. For the 4B-parameter model, validation and checkpointing are performed every 200 steps with patience 5. The protocol-selection experiments with the 270M-parameter model validate and checkpoint every 150 steps with the same patience. Early stopping is based on validation loss, and the checkpoint with the lowest validation loss is restored for subsequent training rather than retaining the final training step.

Matched no-Stage-A configurations start at Stage B rather than from a Stage A checkpoint to assess the initial adaptation stage's contribution. Stage B settings and data composition are matched.

5.3.2 Stage B: Targeted Variety Adaptation

Stage B shifts training from broad subtitle supervision toward more targeted EP/BP resources. It uses the reviewed GPT-Wikipedia resource described in Section 4.2 as its primary data source. A second Stage B condition incorporates the complete FRMT training split, allowing the effect of data composition to be evaluated while keeping the same task templates and training procedure. The task composition of

both Stage B conditions is summarized in Table 5.2. Adding FRMT increases the amount of both rewriting and identification supervision and shifts the mixture slightly toward identification examples.

The GPT-Wikipedia + FRMT data mix concatenates the corresponding GPT-Wikipedia and FRMT training splits and, separately, their validation splits. The mix uses no weighting or resampling.

The 4B-parameter model uses batch size 1, gradient accumulation 32, a peak learning rate of 10^{-5}, 150 warm-up steps, and a maximum of 1200 training steps. The protocol-selection experiments with the 270M-parameter model use batch size 1, gradient accumulation 16, a peak learning rate of 2×10^{-5}, 150 warm-up steps, and a fixed budget of 1200 training steps. If the complete training budget is reached, these settings correspond to at most 38,400 training-example presentations for the 4B-parameter model and 19,200 for the 270M-parameter model. For the 4B-parameter model, both Stage B training sets are smaller than the 38,400-presentation budget, so examples can repeat across shuffled epochs. For the 270M-parameter model, the 19,200-presentation budget exceeds the GPT-Wikipedia training set but not the GPT-Wikipedia + FRMT data mix.

Each Stage B condition uses a held-out validation split constructed with the same task templates as its training split. Matched encoder-controlled and decoder-controlled configurations within the same data condition use the same underlying training and validation examples. For the 4B-parameter model, validation and checkpointing are performed every 100 steps with patience 3. The protocol-selection experiments with the 270M-parameter model also validate and checkpoint every 100 steps, using patience 5. Early stopping is based on validation loss, and the checkpoint with the lowest validation loss is restored.

A secondary decoder-controlled ablation removes equal source-reference rewriting examples from the supervised data while retaining identification examples. The corresponding Stage C variant uses the same continuation data, with aggregate results reported in Appendix E.

5.3.3 Stage C: Reward-Based Fine-Tuning

Stage C applies reward-based continuation to the corresponding Stage B configurations using more conservative rewriting-only data. The GPT-Wikipedia configuration starts from the model trained with GPT-Wikipedia data, while the GPT-Wikipedia + FRMT configuration starts from the model trained with the corresponding data mix. Stage C is not applied to the no-Stage-A configurations.

To select conservative rewriting examples for Stage C, several measures are computed over each aligned EP/BP pair to quantify the extent and type of the differences between the two sentences. A case-insensitive sequence alignment over word and punctuation tokens is used to identify changed regions. A changed region corresponds to one contiguous non-matching alignment block. The edit proportion measures the number of changed word tokens relative to the larger sentence, while structural similarity is computed from the sequence-alignment ratio over the complete token sequences. Changes are also compared with a predefined inventory of EP/BP marker forms. Changed word tokens in the non-matching alignment blocks are divided into marker and non-marker tokens according to case-insensitive membership in this inventory. The non-marker change proportion is the number of non-marker changed-token occurrences across both sentences divided by the larger sentence word count.

For a non-equal pair to be included in the Stage C training set, it must contain at most four changed regions, have an edit proportion of at most 0.30, and have a structural similarity score of at least 0.72. The non-marker change proportion must not exceed 0.18, and the pair may contain at most six non-marker changed-token occurrences. When at least one marker token changes, the number of non-marker changed tokens may exceed the number of marker changed tokens by at most two. This final constraint is not applied when no marker token changes. Unchanged pairs are retained separately as no-change examples, for which copying the source is the correct output.

Both Stage C training sets are balanced across rewriting directions. The combined set contains the same 6,686 GPT-Wikipedia rewriting examples used in the GPT-Wikipedia set together with 142 FRMT examples derived from 71 aligned pairs. No-change examples make up about 35% of each Stage C set.

Stage C contains no identification examples. Under decoder control, rewriting targets still begin with `PT` or `BR`, and correctness of this first generated token contributes to the reward.

The Stage C configurations with the 4B-parameter model use batch size 1, gradient accumulation 8, a learning rate of 5×10^{-6}, 20 warm-up steps, and a fixed optimization budget of 600 steps. AdamW is used with weight decay 0 and the same linear learning-rate schedule defined for the preceding stages. Unlike the supervised stages, Stage C does not use validation-based early stopping or best-validation-loss checkpoint selection. The configurations use seed 42 and gradient clipping with maximum norm 1.0. The final training state after 600 steps is used for evaluation.

The optimization uses a group-relative update inspired by the reward optimization methods described in Section 2.5 (Shao et al., 2024). For each training input, four candidate rewrites are sampled with temperature 0.8, top-p 0.95, a maximum completion length of 128 tokens, and no n-gram blocking. The gold target is then included in the same candidate group, so each group usually contains five candidate entries. The gold target participates in reward normalization like the sampled candidates.

The reward compares each rewrite with the reference and unchanged source. For the BLEU component, let B_m denote candidate sentence-level BLEU and B_c the BLEU from copying the source:

$$R_{\mathrm{BLEU}} = \frac{B_m}{B_m + B_c + \epsilon}. \tag{5.1}$$

Similarly, let T_m be candidate TER and T_c the copy-source TER:

$$R_{\mathrm{TER}} = \frac{T_c + \epsilon}{T_m + T_c + 2\epsilon}. \tag{5.2}$$

The constant $\epsilon = 10^{-8}$ prevents division by zero.

These forms reward a candidate relative to the unchanged-source alternative rather than using the raw BLEU and TER values alone. Encoder-controlled models weight the two components equally:

$$R_{\mathrm{enc}} = 0.5R_{\mathrm{BLEU}} + 0.5R_{\mathrm{TER}}, \tag{5.3}$$

Decoder-controlled models additionally reward correctness of the first generated input-variety label:

$$R_{\text{dec}} = 0.4R_{\text{BLEU}} + 0.4R_{\text{TER}} + 0.2R_{\text{label}}. \tag{5.4}$$

For BP-to-EP examples, R_{label} is 1 when the first generated label is `BR`. For EP-to-BP examples, it is 1 when the first generated label is `PT`. Otherwise, it is 0. After removal of any decoder control label and whitespace normalization, 0.10 is subtracted from the combined candidate reward when the generated rewrite exactly matches the source and the reference requires a change.

Candidate rewards are converted to group-normalized advantages using the procedure described in Section 2.5. If the reward standard deviation within a group is zero, all advantages for that group are set to zero. The implemented Stage C loss is

$$\mathcal{L}_{\text{StageC}}(\theta) = -\frac{1}{|G(x)|} \sum_{y_i \in G(x)} A_i \frac{\log p_\theta(y_i \mid x)}{L_{\max}}, \tag{5.5}$$

where A_i is the group-normalized advantage for candidate y_i and $L_{\max} = 128$ is the maximum completion length used for normalization. The implemented objective does not use Proximal Policy Optimization (PPO)-style likelihood ratios or clipping. At $\beta = 0$, the starting policy has no KL penalty.

Stage C differs from Stage B in both the optimization procedure and the training data. Thus, Stage B-to-Stage C comparisons characterize the Stage C procedure rather than the reward objective alone.

5.4 Supervised Protocol Selection

The protocol-selection study with the 270M-parameter model compares four decoder-controlled supervised-training configurations at lower computational cost before the experiments with the 4B-parameter model. All configurations use full fine-tuning and follow the same Stage A and Stage B training path described in Section 5.3, with OpenSubtitles used in Stage A and GPT-Wikipedia used in Stage B. The comparison keeps the same T5Gemma 2 checkpoint, decoder-controlled task formulation, task construction, evaluation procedure described in Section 6.1, and comparable step-budget settings, while varying only the supervised protocol component being tested. The results are reported in Section 6.2.

The comparison comprises one base protocol and three single-factor variants: task-balanced training, added-token treatment, and restricted classification loss. In the base protocol, rewriting and identification rows are combined in the same training mixture and shuffled together, so their contribution follows their relative frequency in the data rather than an explicit task balance. The control strings use the existing base-vocabulary representations of `PT` and `BR` described in Section 5.2.2. Classification examples are scored against the normal decoder vocabulary, with the supervised target positions determined by the formulation described in Section 5.2.

The task-balanced variant changes only the composition of the examples contributing to each optimizer update, enforcing equal numbers of rewriting and identification rows. The data, task templates, tokenizer treatment, and identification loss remain unchanged from the base protocol.

The added-token variant keeps the same control strings and task templates but explicitly registers them with the tokenizer instead of using their existing base-vocabulary representations. The data mixture, sampling procedure, and loss computation remain unchanged from the base protocol.

The restricted-classification-loss variant changes only classification loss by restricting scoring to `PT` and `BR`. Rewriting examples still use the full decoder vocabulary.

6

Experiments and Evaluation

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This chapter evaluates the trained models on variety identification and bidirectional rewriting. Section 6.1 defines the evaluation setup, and Section 6.2 reports the protocol-selection study with the 270M-parameter model used to choose the supervised training protocol. Sections 6.3 and 6.4 analyze identification and rewriting with the 4B-parameter model, while Section 6.5 examines copying behavior, representative outputs, and limitations of the available supervision.

6.1 Evaluation Setup

Evaluation uses the FRMT test split and the Golden Collection described in Chapter 4. Golden Collection evaluation uses the manually curated reference instead of the automatic DeepL reference.

Rewriting quality is measured with BLEU (Papineni et al., 2002), computed with SacreBLEU (Post, 2018), and TER (Snover et al., 2006), with higher BLEU and lower TER indicating closer agreement with the reference. BLEU values use the 0–100 reporting scale, and rewriting uses beam search with beam size 4. Decoder-side variety labels are removed before scoring. Rewriting comparisons are reported separately for BP-to-EP and EP-to-BP when directional results are available, while the Stage A ablation uses aggregate rewriting metrics. For evaluation, corpus BLEU and TER are used

for aggregate model comparisons, while sentence-level BLEU and TER are reserved for the diagnostic analyses in Section 6.5. Stage C separately uses sentence-level BLEU and TER within the training reward described in Section 5.3.3. Corpus BLEU is computed over each complete dataset-and-direction slice by aggregating candidate n-gram counts and the corresponding reference-limited match counts across all sentences before calculating the corpus-level score.

Many EP/BP examples require only localized changes, and some require no modification. The evaluation therefore also includes a deterministic copy baseline that returns the source sentence unchanged, providing a reference for how favorable each evaluation set is to source preservation.

Variety identification is evaluated with accuracy and macro-F1. Predictions are obtained by scoring `PT` and `BR` under the adapted sequence-to-sequence model and selecting the higher-scoring label.

6.2 Supervised Protocol Selection with the 270M-Parameter T5Gemma 2 Model

The supervised protocol was selected with the fully fine-tuned 270M-parameter T5Gemma 2 model under decoder control so that alternative training choices could be compared at lower computational cost. Table 6.1 reports the base protocol, task-balanced batches, added-token treatment, and restricted classification loss after the same OpenSubtitles Stage A and GPT-Wikipedia Stage B path. The variants differ only in the protocol component defined in Section 5.4.

Table 6.1: Decoder-controlled 270M full-fine-tuning supervised protocol-selection ablations.

Protocol variant	FRMT					Golden Collection				
	Macro-F1 (%)	BLEU (0–100)		TER		Macro-F1 (%)	BLEU (0–100)		TER	
		BP-to-EP	EP-to-BP	BP-to-EP	EP-to-BP		BP-to-EP	EP-to-BP	BP-to-EP	EP-to-BP
Base protocol	70.76	38.56	38.69	**0.4748**	**0.4828**	67.63	86.45	86.70	0.0978	0.0981
Task-balanced batches	69.74	38.56	38.55	0.4751	0.4859	68.52	86.04	86.45	0.0987	0.1001
Added-token path	**72.93**	38.36	38.67	0.4768	0.4864	66.66	86.23	86.60	0.0986	0.1006
Restricted classification loss	70.74	**38.70**	**38.72**	0.4750	0.4832	**69.10**	**87.01**	**87.32**	**0.0941**	**0.0956**

Task-balanced batches do not provide a broad improvement: they slightly lower FRMT identification and rewriting, improve Golden Collection identification, and slightly lower Golden Collection rewriting. The added-token path gives the largest FRMT identification gain, but slightly lowers FRMT rewriting, lowers Golden Collection identification, and does not improve Golden Collection rewriting. Restricted classification loss gives the strongest Golden Collection identification and rewriting values, and only a small FRMT BLEU increase, with a specialized classification loss but no FRMT identification gain.

Although individual variants improve particular metrics, none provides a consistent gain across identification and rewriting. The base protocol is therefore retained as the simpler configuration for the subsequent experiments with the 4B-parameter model. The retained protocol uses the natural rewriting-to-identification mixture, the existing `PT`/`BR` base-vocabulary representations, and normal decoder-vocabulary scoring at the supervised label position, with the complete target sequence used for rewriting loss.

6.3 Variety Identification Results

Identification performance varies more clearly with the Stage B supervision condition than with control placement, particularly on FRMT. Table 6.2 reports these differences and the Stage C results.

Table 6.2: Identification results for the 4B-parameter T5Gemma 2 model with rank-48 LoRA.

Stage	Control	Data condition	FRMT		Golden Collection	
			Accuracy (%)	Macro-F1 (%)	Accuracy (%)	Macro-F1 (%)
B	Encoder	GPT-Wikipedia	82.39	82.39	**73.99**	**73.92**
B	Decoder	GPT-Wikipedia	81.69	81.68	71.61	71.53
B	Encoder	GPT-Wikipedia + FRMT	86.09	86.06	71.43	71.19
B	Decoder	GPT-Wikipedia + FRMT	**86.15**	**86.10**	72.53	72.38
C	Encoder	GPT-Wikipedia	82.33	82.33	73.63	73.50
C	Decoder	GPT-Wikipedia	81.52	81.31	72.34	71.81
C	Encoder	GPT-Wikipedia + FRMT	85.82	85.78	71.43	71.17
C	Decoder	GPT-Wikipedia + FRMT	85.65	85.54	72.34	72.02

Adding FRMT supervision improves FRMT identification under both control formulations, with macro-F1 increasing from 82.39 to 86.06 under encoder control and from 81.68 to 86.10 under decoder control. The effect differs on the Golden Collection: encoder-controlled macro-F1 decreases from 73.92 to 71.19, while decoder-controlled macro-F1 increases from 71.53 to 72.38. Control placement also changes with the supervision condition. With GPT-Wikipedia alone, encoder control gives the higher macro-F1 on both evaluation resources, whereas after adding FRMT, decoder control is slightly higher on both.

Accuracy and macro-F1 remain very close across all configurations, with identification evaluated on equal numbers of PT- and BR-labeled examples.

The FRMT results also allow a comparison with the identification results reported by Sousa et al. (2025b). The Stage B encoder-controlled configuration trained on GPT-Wikipedia reaches 82.39 macro-F1 on FRMT, compared with the 77.25 F1 of their strongest BERTimbau-based identifier. Although both studies use FRMT for evaluation, their training setups and objectives differ: the previous model is dedicated to identification, whereas this work uses a shared encoder-decoder interface.

Identification changes only slightly from Stage B to the corresponding Stage C configurations. Across the four matched comparisons and both evaluation resources, the largest absolute macro-F1 change is 0.56 percentage points, from 86.10 to 85.54 for the decoder-controlled GPT-Wikipedia + FRMT configuration on FRMT. The remaining changes are smaller and include both increases and decreases, with no consistent identification improvement or systematic degradation across the evaluated conditions.

6.4 Variety Rewriting Results

Rewriting is more sensitive than identification to changes in supervision composition. Table 6.3 summarizes performance across supervision settings, control formulations, and training stages.

Adding FRMT supervision has a direction-dependent effect on FRMT rewriting, ranging from almost no change to a clear improvement. In contrast, Golden Collection BLEU drops by roughly 20 points across every direction and control comparison, with corresponding increases in TER.

Table 6.3: Directional Stage B/C rewriting results for the 4B-parameter T5Gemma 2 model with rank-48 LoRA.

Stage	Control	Data condition	FRMT		Golden Collection	
			BP-to-EP BLEU (0–100) / TER	EP-to-BP BLEU (0–100) / TER	BP-to-EP BLEU (0–100) / TER	EP-to-BP BLEU (0–100) / TER
B	Encoder	GPT-Wikipedia	37.87/0.4791	40.10/0.4714	86.08/0.0946	**89.09**/**0.0823**
B	Encoder (rewriting-only)	GPT-Wikipedia	37.73/0.4803	40.31/0.4715	85.66/0.0976	88.66/0.0847
B	Decoder	GPT-Wikipedia	40.45/0.4594	40.60/0.4673	87.75/0.0874	87.50/0.0916
B	Encoder	GPT-Wikipedia + FRMT	37.99/0.4770	**45.43**/0.4273	65.14/0.2555	66.61/0.2535
B	Encoder (rewriting-only)	GPT-Wikipedia + FRMT	36.76/0.4852	45.40/**0.4251**	56.10/0.3222	58.69/0.3103
B	Decoder	GPT-Wikipedia + FRMT	**43.97**/**0.4338**	44.74/0.4306	68.37/0.2336	67.83/0.2400
C	Encoder	GPT-Wikipedia	37.89/0.4784	40.28/0.4696	86.22/0.0953	88.93/0.0837
C	Decoder	GPT-Wikipedia	40.34/0.4603	40.33/0.4698	**88.44**/**0.0824**	88.02/0.0878
C	Encoder	GPT-Wikipedia + FRMT	38.35/0.4756	43.47/0.4440	77.16/0.1601	79.95/0.1488
C	Decoder	GPT-Wikipedia + FRMT	42.72/0.4417	42.72/0.4466	81.52/0.1351	80.70/0.1402

To examine whether identification supervision changes rewriting quality, the encoder-controlled Stage B comparison includes matched multitask and rewriting-only configurations trained with the same rewriting data. With GPT-Wikipedia, removing identification supervision changes rewriting by less than half a BLEU point in every comparison. With GPT-Wikipedia + FRMT, the two configurations remain close on FRMT, while the multitask configuration is stronger on the Golden Collection by 9.04 BLEU in BP-to-EP and 7.92 BLEU in EP-to-BP. Removing identification supervision does not improve encoder-controlled rewriting.

The effect of control placement depends on rewriting direction. Decoder control performs better in every BP-to-EP comparison, while EP-to-BP is mixed and sometimes favors encoder control.

Beyond these supervised comparisons, Stage C shows how the models change after reward-based continuation. Reward-based continuation changes the GPT-Wikipedia models only slightly. The larger effect appears after GPT-Wikipedia + FRMT specialization, where Golden Collection BLEU recovers by roughly 12–13 points across the direction and control comparisons. Changes in FRMT rewriting performance remain comparatively small and mixed. Stage C therefore recovers much of the Golden Collection performance lost after Stage B specialization with the GPT-Wikipedia + FRMT data mix, while its FRMT effect remains direction- and formulation-dependent.

The most directly comparable prior rewriting result is BP-to-EP evaluation on the Golden Collection. After rescaling the BLEU values reported by Sanches et al. (2024) to the 0–100 scale, their strongest trained system obtains approximately 84 BLEU and 0.10 TER, while their published single-direction BP-to-EP copy baseline reaches approximately 89 BLEU and 0.07 TER. The decoder-controlled GPT-Wikipedia configuration reaches 87.75 BLEU and 0.0874 TER at Stage B and 88.44 BLEU and 0.0824 TER after Stage C. This numerical relation is contextual, as Sanches et al. (2024) train specifically for BP-to-EP rewriting, while the present configuration is bidirectional and also supports identification.

Additional descriptive comparisons with external pretrained and instruction-following models evaluated under the same protocol are reported in Appendix D.

The Stage A ablation reported in Table 6.4 compares each configuration with a matched model that starts directly from Stage B. To capture effects on both tasks, the comparison reports macro-F1 together with aggregate rewriting BLEU and TER deltas.

Table 6.4: Stage A warm-up ablation under aggregate evaluation. Deltas are computed as the Stage A model minus the matched no-Stage-A model. Positive deltas indicate improvement for macro-F1 and BLEU, while negative deltas indicate improvement for TER.

Control	Data condition	Dataset	△ Macro-F1 (pp)	△ BLEU (points)	△ TER
Encoder	GPT-Wikipedia	FRMT	**+0.87**	**+4.40**	**-0.0573**
Encoder	GPT-Wikipedia	Golden Collection	**+4.01**	**+25.41**	**-0.2404**
Encoder	GPT-Wikipedia + FRMT	FRMT	+0.81	+1.92	-0.0176
Encoder	GPT-Wikipedia + FRMT	Golden Collection	+1.62	-6.79	+0.0425
Decoder	GPT-Wikipedia	FRMT	-1.28	-0.01	+0.0004
Decoder	GPT-Wikipedia	Golden Collection	+2.67	+1.42	-0.0144
Decoder	GPT-Wikipedia + FRMT	FRMT	-0.07	+0.35	-0.0031
Decoder	GPT-Wikipedia + FRMT	Golden Collection	+2.19	+2.00	-0.0217

For configurations whose Stage B uses GPT-Wikipedia, the clearest aggregate Stage A benefit occurs on the Golden Collection, especially under encoder control, where Stage A adds 4.01 macro-F1 points and 25.41 BLEU while reducing TER by 0.2404. The decoder-controlled GPT-Wikipedia condition also improves on the Golden Collection, while the GPT-Wikipedia + FRMT conditions show smaller or mixed changes.

Overall, supervision composition produces the clearest changes in rewriting performance. GPT-Wikipedia alone is consistently better suited to the Golden Collection at Stage B, while adding FRMT can improve FRMT rewriting depending on direction and control formulation but sharply reduces Golden Collection performance. Decoder control shows the stronger overall tendency for rewriting, performing better in every BP-to-EP comparison, although EP-to-BP remains mixed. Stage A generally helps the training process, with its largest gains in configurations whose Stage B uses GPT-Wikipedia, while Stage C produces its largest gains after GPT-Wikipedia + FRMT specialization, where it consistently recovers Golden Collection performance across control formulations and rewriting directions.

6.5 Error Analysis

Section 6.4 shows that adding FRMT supervision can improve FRMT rewriting while sharply reducing Golden Collection performance at Stage B. To examine this contrast, Section 6.5.1 first considers how closely references remain to their sources, Section 6.5.2 then compares the copying and rewriting behavior under the two Stage B conditions, and Section 6.5.3 examines supervision limitations.

6.5.1 Copy-Friendly Evaluation Cases

The Golden Collection contains many cases in which little or no rewriting is required. Of its 1,000 directional cases, 454 have identical source and reference sentences. When the source itself is used as the candidate, 566 cases obtain TER at most 0.05 and 636 obtain TER at most 0.08. At corpus level, the unchanged-source baseline reaches 89.17 BLEU and 0.0755 TER on the Golden Collection, compared with 38.75 BLEU and 0.4819 TER on FRMT. The Golden Collection is therefore much more favorable to conservative outputs, whereas FRMT references differ more substantially from their sources.

Table 6.5 compares sentence-level BLEU (sBLEU) and TER with the unchanged-source baseline.

Table 6.5: Golden Collection sentence-level predictions matching or outperforming the unchanged-source baseline for the Stage B models in Table 6.3.

Data condition	Control	Direction	sBLEU $\geq$ copy (%)	TER $\leq$ copy (%)
GPT-Wikipedia	Encoder	BP-to-EP	78.6	78.8
GPT-Wikipedia	Encoder	EP-to-BP	81.4	81.2
GPT-Wikipedia	Decoder	BP-to-EP	74.0	74.4
GPT-Wikipedia	Decoder	EP-to-BP	72.8	73.6
GPT-Wikipedia + FRMT	Encoder	BP-to-EP	23.0	23.0
GPT-Wikipedia + FRMT	Encoder	EP-to-BP	23.6	24.4
GPT-Wikipedia + FRMT	Decoder	BP-to-EP	31.0	31.2
GPT-Wikipedia + FRMT	Decoder	EP-to-BP	30.2	29.8

The sentence-level comparison in Table 6.5 counts a prediction when its sentence-level BLEU matches or exceeds the copy baseline, or when its sentence-level TER matches or improves on the copy baseline. Across both control formulations, predictions under GPT-Wikipedia supervision match or outperform copying in 72.8–81.4% of cases by sentence-level BLEU and 73.6–81.2% by sentence-level TER. With GPT-Wikipedia + FRMT, the corresponding ranges fall to 23.0–31.0% and 23.0–31.2%, respectively.

Sentence-level comparisons cannot be read as sentence-weighted versions of the corpus metrics. Corpus BLEU aggregates n-gram match statistics over the complete evaluation slice instead of averaging sentence-level BLEU or counting wins. A model can therefore match or improve over copying on a large proportion of individual sentences yet obtain lower corpus BLEU when losses on other examples remove proportionally more matched n-grams. This sensitivity is especially relevant for the many short, copy-friendly Golden Collection sentences. Changing one word in a short sentence can disrupt a large fraction of its available n-grams, so an unnecessary change can incur a disproportionate overlap loss. A correct small edit and an unnecessary small edit can consequently have very different effects depending on the reference. Corpus TER similarly sums edit counts over the complete evaluation slice and divides by the total reference-token count. It does not average sentence-level TER values, so sentence-level gains can coexist with worse corpus TER when losses elsewhere add more edits per reference token.

Table 6.6 illustrates both outcomes with one conservative Stage B GPT-Wikipedia checkpoint. In the first and second rows, the model applies the required local substitution and matches the reference exactly. In the third and fourth rows, copying is correct, but rewriting sharply lowers sentence-level BLEU.

Table 6.6: Golden Collection examples comparing the unchanged-source baseline with predictions from a conservative Stage B GPT-Wikipedia checkpoint. Boldface marks the sentence spans responsible for the illustrated contrast, and the higher sBLEU value in each row is bold.

Source	Reference	Model prediction	Copy sBLEU (0–100)	Model sBLEU (0–100)
*Clonar animais continua **sendo** uma tarefa difícil.*	*Clonar animais continua **a ser** uma tarefa difícil.*	*Clonar animais continua **a ser** uma tarefa difícil.*	44.12	**100.00**
*O artigo do **investigador** está na edição de hoje da revista "Science".*	*O artigo do **pesquisador** está na edição de hoje da revista "Science".*	*O artigo do **pesquisador** está na edição de hoje da revista "Science".*	80.03	**100.00**
***V.Ex.ª** está inscrito para falar contra.*	***V.Ex.ª** está inscrito para falar contra.*	***Sua Excelência** está inscrito para falar contra.*	**100.00**	46.76
— Ah!	*— Ah!*	***É...***	**100.00**	0.00

The much weaker FRMT copy baseline reflects references that are farther from their sources on average. Broader rewriting can therefore improve corpus-level agreement more readily than on the Golden Collection, although broader changes are not necessarily appropriate for every example in FRMT.

6.5.2 Effect of FRMT During Supervised Specialization

Given the Golden Collection's sensitivity to unnecessary changes, the next question is whether the two Stage B supervision conditions differ in how often they preserve the source. The encoder-controlled Stage B models trained with GPT-Wikipedia data and with the GPT-Wikipedia + FRMT data mix differ clearly in this behavior. On the 454 exact-copy Golden Collection rows, where leaving the source unchanged matches the reference, the model trained with GPT-Wikipedia preserves the source exactly in 74.7% of cases, compared with 18.7% for the model trained with the GPT-Wikipedia + FRMT data mix. On the 546 non-exact rows, the corresponding copy rates are 33.7% and 3.1%. In these cases, copying can indicate under-editing, so reducing it can realize required changes but also encourage unnecessary reformulation.

Table 6.7 illustrates both sides of this trade-off. In the first row, the model trained with GPT-Wikipedia under-edits by leaving *procura* unchanged, whereas the model trained with the GPT-Wikipedia + FRMT data mix makes the required change to *demanda* but also reformulates the sentence more broadly. In the fourth row, the reference is identical to the source, which the GPT-Wikipedia model preserves, while the GPT-Wikipedia + FRMT model rewrites unnecessarily.

Table 6.7: Illustrative Golden Collection predictions from encoder-controlled Stage B models trained with GPT-Wikipedia and GPT-Wikipedia + FRMT. Boldface marks model-output spans that illustrate under-editing or unnecessary reformulation discussed in the text.

Source	Reference	GPT-Wikipedia	GPT-Wikipedia + FRMT
"A procura aumenta de forma drástica no mundo todo", afirmou o especialista em recursos hídricos José Galizia Tundisi, do Instituto Internacional de Ecologia, em São Carlos (SP). *Ele afirmou que o aumento de temperatura, que pode chegar a 3C nas áreas húmidas, seria um desastre.* *Segundo o investigador, a equipa inclusive pediu patentes sobre algumas dessas aplicações, mas, enquanto o pedido não for aprovado, Perales prefere não revelar exatamente do que se trata.* *Daí por que está absolutamente correto no conteúdo, como na forma, o parecer do ilustre Relator Bernardo Cabral.*	*"A demanda aumenta de forma drástica no mundo todo", afirmou o especialista em recursos hídricos José Galizia Tundisi, do Instituto Internacional de Ecologia, em São Carlos (SP).* *Ele afirmou que o aumento de temperatura, que pode chegar a 3C nas áreas úmidas, seria um desastre.* *Segundo o pesquisador, a equipe inclusive pediu patentes sobre algumas dessas aplicações, mas, enquanto o pedido não for aprovado, Perales prefere não revelar exatamente do que se trata.* *Daí por que está absolutamente correto no conteúdo, como na forma, o parecer do ilustre Relator Bernardo Cabral.*	*"A **procura** aumenta de forma drástica no mundo todo", afirmou o especialista em recursos hídricos José Galizia Tundisi, do Instituto Internacional de Ecologia, em São Carlos (SP).* *Ele afirmou que o aumento de temperatura, que pode chegar a 3C nas áreas úmidas, seria um desastre.* *Segundo o pesquisador, a equipe inclusive pediu patentes sobre algumas dessas aplicações, mas, enquanto o pedido não for aprovado, Perales prefere não revelar exatamente do que se trata.* *Daí por que está absolutamente correto no conteúdo, como na forma, o parecer do ilustre Relator Bernardo Cabral.*	*"A demanda **está aumentando drasticamente em todo o mundo**", **disse** o especialista em recursos hídricos José Galizia Tundisi, do Instituto Internacional de Ecologia em São Carlos (SP).* *Ele **disse** que o aumento de temperatura, que pode chegar a **3ºC em áreas úmidas**, seria um desastre.* *Segundo o pesquisador, a equipe **até mesmo solicitou** patentes sobre algumas dessas aplicações, mas enquanto o pedido não for aprovado, Perales prefere não revelar exatamente do que se trata.* ***Por isso, o parecer do ilustre Relator, Bernardo Cabral, é absolutamente correto tanto no conteúdo como na forma.***

6.5.3 Limitations of the Available Supervision

The broader rewriting behavior observed after adding FRMT supervision may reflect differences in the aligned data itself. Inspection of the available pairs shows source–reference differences that extend beyond localized EP/BP conversion, including lexical reformulation and changes in expressed information. In some FRMT examples, lexical substitutions such as *marcante* to *importante*, *ganhou* to *recebeu*, *reside* to *vive*, *usar* to *utilizar*, and *música* to *canção* alter the wording without a clear EP/BP adaptation.

A second issue concerns differences in the information expressed by the source and reference. Table 6.8 shows representative cases from both evaluation resources. In the Golden Collection examples, the reference removes or compresses source material, while in the FRMT examples it removes repeated title glosses and unit conversions. These transformations go beyond direct language-variety substitution. Additional inspected FRMT examples contain other omissions or editorial changes of the same general kind, so the cases in Table 6.8 are illustrative rather than exhaustive.

Table 6.8: Examples where source and reference sentences differ in expressed information. Boldface marks the spans responsible for the information difference discussed in the text.

Dataset	Direction	Source	Reference
Golden Collection	BP-to-EP	*- Oww você é um amor Obrigado viu!- Falei* ***e o mesmo retribuiu com um abraço.***	*- Oww, és adorável. Obrigado!*
Golden Collection	EP-to-BP	*Então, o latifundiário pode tirar toda a madeira, expulsar os* ***que ocuparam a terra abandonada*** *e fazer verdadeiros* ***atos*** *absurdos nesse período.*	*Então, o latifundiário pode tirar toda a madeira, expulsar os* ***posseiros*** *e fazer verdadeiros absurdos nesse período.*
FRMT	BP-to-EP	*A UNICEF concedeu a Lerner o Prêmio Criança e Paz em 1996 pelos seus programas "Da Rua para a Escola"* ***(Da Rua para a Escola)****, "Protegendo a Vida"* ***(Protegendo a Vida)*** *e "Universidade do Professor"* ***(Universidade do Professor)****.*	*A UNICEF atribuiu a Lerner o Prémio Criança e Paz em 1996 pelos programas "Da Rua para a Escola", "Protegendo a Vida" e "Universidade do Professor".*
FRMT	BP-to-EP	*O Autódromo Internacional Orlando Moura tem uma pista de 3.433 metros* ***(11.263 pés)****, e o Kartódromo Ayrton Senna uma pista de 930 metros* ***(3.051 pés)****.*	*O Autódromo Internacional Orlando Moura tem uma pista de 3433 metros e o Kartódromo Ayrton Senna tem uma pista de 930 metros.*

Source–reference differences beyond localized EP/BP conversion can affect both training and evaluation. In supervision, they can teach transformations broader than controlled EP/BP rewriting, while in evaluation references they can affect scores even for meaning-preserving, appropriate variety-specific changes.

BLEU and TER compare a model output with a particular reference, so they cannot determine whether a difference from that reference reflects a necessary variety-specific edit, an acceptable alternative, or an unnecessary reformulation. This ambiguity becomes more important when the reference itself contains paraphrastic or editorial changes. The same issue also carries into Stage C, where sentence-level BLEU and TER contribute directly to the reward. If a reference contains a broader rewrite than the task requires, the reward can favor candidates that reproduce that rewrite over candidates that make a smaller but still appropriate variety-specific change. Conversely, when copying is appropriate, the relative-to-copy formulation can reward restraint.

7

Conclusion

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This chapter brings together the main findings from the identification and rewriting experiments. Section 7.1 summarizes the main empirical conclusions and their implications for Portuguese variety adaptation, while Section 7.2 outlines future work on data, evaluation, and staged training.

7.1 Summary

This work studied European and Brazilian Portuguese through variety identification and bidirectional rewriting in a shared T5Gemma 2 encoder-decoder framework. Lower-cost experiments with the 270M-parameter model were used to select the supervised protocol and examine reward-design choices, while the 4B-parameter model was used for the main comparisons of control formulation, supervision composition, and staged adaptation. The protocol-selection experiments improved individual metrics under some variants, but none consistently improved performance across both tasks and evaluation resources.

The clearest result from the 4B-parameter experiments concerns supervision composition. GPT-Wikipedia produces substantially better Golden Collection rewriting than the GPT-Wikipedia + FRMT data mix, while the effect of adding FRMT on FRMT itself depends on rewriting direction and control formulation. The identification task is less sensitive to this change. Adding FRMT improves FRMT identification under both formulations, while its effect on the Golden Collection is mixed. The performance differences between encoder and decoder control also depend on rewriting direction. Decoder control performs better in every

BP-to-EP rewriting comparison, whereas EP-to-BP includes cases where encoder control performs better. Removing identification supervision from the encoder-controlled models does not improve rewriting overall.

The reported models also perform better numerically than the closest published systems used for comparison. The encoder-controlled GPT-Wikipedia model exceeds the published FRMT identification result reported by Sousa et al. (2025b), while the decoder-controlled GPT-Wikipedia model exceeds the strongest trained BP-to-EP Golden Collection result reported by Sanches et al. (2024). These comparisons remain contextual, since the previous systems use different training setups and objectives.

For configurations whose Stage B uses GPT-Wikipedia, Stage A produces its largest gains on the Golden Collection. Stage C changes the GPT-Wikipedia conditions only modestly. However, after GPT-Wikipedia + FRMT specialization, Stage C recovers a substantial part of the Golden Collection performance lost at Stage B, while its effect on FRMT rewriting remains mixed. The identification results remain close to the corresponding Stage B values after Stage C despite the continuation stage using rewriting-only data.

Many Golden Collection references require little or no change, making conservative outputs highly competitive. Models trained with GPT-Wikipedia preserve the source much more often than models trained with the GPT-Wikipedia + FRMT data mix. Adding FRMT can reduce under-editing when a change is required, but it also increases broader reformulation. Inspection of the aligned data shows that some FRMT pairs contain lexical substitutions, omissions, and other changes in expressed information that go beyond the localized changes required for EP/BP rewriting. Similar reference-side differences also occur in some Golden Collection examples.

These reference properties also affect what can be concluded from automatic evaluation. BLEU and TER measure agreement with a particular reference, but that reference does not always isolate the variety-specific change required by the task. The same issue affects Stage C when sentence-level BLEU and TER contribute to the reward, since a broader-than-required reference can favor candidates that reproduce that broader rewrite. Across the experiments, supervision composition produces the largest changes in rewriting behavior, while the error analysis shows that the construction of the aligned pairs influences both the behavior learned by the models and how that behavior is evaluated.

7.2 Future Work

The strongest direction for future work is to improve the supervision used for controlled rewriting. Larger manually verified resources would make it possible to distinguish required variety-specific changes from optional paraphrase and changes in expressed information more reliably. The OpenSubtitles filtering procedure could also be evaluated more systematically by varying the SimAlign coverage threshold and allowed sentence-length ratio, and by comparing the current XLM-R, word-level intersection setup with alternative encoders, alignment granularities, and extraction methods. For Stage C data selection, the existing structural-similarity and edit criteria could be supplemented with content-preservation checks for entities, numbers, parenthetical information, units, and other source–reference differences. This could remove pairs with differences beyond the requested variety change despite high structural similarity.

Improved supervision should be accompanied by evaluation criteria that are less dependent on a single reference. Future evaluation should assess whether the required variety-specific changes were made while preserving the source meaning and avoiding unnecessary rewriting. Human evaluation would provide the strongest basis for validating these judgments, while targeted linguistic checks, multiple acceptable references, or learned evaluators could support larger-scale evaluation.

A validated evaluation signal would also make it possible to study Stage C under better controlled conditions. The current Stage C comparison changes both the optimization procedure and the training subset, so a follow-up should vary these factors independently. One comparison could keep the Stage C training subset fixed and contrast supervised continuation with reward-based optimization, while another could hold the optimization method fixed and vary the training subset. Once these effects are separated, reward design could be studied more directly by comparing individual quality and conservatism components, explicitly including the unchanged source among the candidates, varying candidate-group construction, or assigning feedback to the particular spans that require adaptation. These experiments could also test whether Stage C identification remains stable under stronger rewriting-focused optimization.

Finally, the stability of these findings should be tested beyond the present resources and T5Gemma 2 configurations. Additional domains, model families, and larger manually evaluated test sets would show whether the observed effects of supervision composition, control formulation, and staged adaptation generalize beyond the current experimental setting. This broader validation would be most informative after supervision and evaluation are better controlled, since it would help separate model-dependent differences from effects caused by the available training and evaluation resources. The descriptive external comparison in Appendix D could also be strengthened by evaluating a current frontier proprietary model under the same identification and bidirectional rewriting protocol.

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A

Additional Data Preparation Details

This appendix provides additional details on the preparation of the resources used in the experiments, focusing on OpenSubtitles and GPT-Wikipedia, whose construction involved substantial preprocessing and filtering. FRMT and the Golden Collection mainly require conversion to the common experimental format and are described in Sections 4.3 and 4.4. This appendix also reports tokenization statistics for the resources used in the experiments.

A.1 Tokenization Statistics

Table A.1 reports corpus token counts and the number of tokenizer vocabulary entries observed with the `google/t5gemma-2-4b-4b` tokenizer used by the 4B-parameter T5Gemma 2 model. These statistics are computed over the complete processed resource splits shown in the table, independently of how many examples are subsequently presented to a model during training. *Total tokens* counts every tokenizer-token occurrence across all sentences in the specified resource variant, whereas *Distinct tokenizer identifiers (IDs)* counts each vocabulary ID only once, regardless of how often it occurs. The large difference between the quantities reported in Table A.1 is therefore expected, since a corpus can contain millions of token occurrences while repeatedly drawing from a much smaller set of vocabulary entries.

A.2 OpenSubtitles

OpenSubtitles is used as a large-scale source of aligned EP/BP sentence pairs. Its main value is coverage: it exposes the models to many lexical contexts, short utterances, informal dialogue, and everyday constructions. The same properties also make it noisy. Subtitle files often contain segmentation artifacts, speaker marks, incomplete fragments, and weak sentence alignment. Preparation therefore

Table A.1: Token and tokenizer-vocabulary statistics for the resources used in the reported experiments.

Resource	Split and variety	Rows	Total tokens	Distinct tokenizer IDs
OpenSubtitles	train, pt-BR	10,347,883	113,121,931	73,773
OpenSubtitles	train, pt-PT	10,347,883	116,213,290	74,179
GPT-Wikipedia	train, pt-BR	4,519	121,589	8,166
GPT-Wikipedia	train, pt-PT	4,519	122,542	8,127
FRMT	train, pt-BR	2,502	94,966	11,044
FRMT	train, pt-PT	2,502	98,345	10,886
FRMT	test, pt-BR	2,611	97,920	11,117
FRMT	test, pt-PT	2,611	101,039	10,874
Golden Collection	test, pt-BR	500	14,166	4,234
Golden Collection	test, pt-PT	500	14,285	4,209

aims to keep as many useful aligned pairs as possible while removing cases where the paired sentences no longer express the same content.

A.2.1 Formatting and Segmentation Repairs

As summarized in Section 4.1, formatting cleanup removes subtitle-specific artifacts such as dialogue dashes, speaker labels, irregular spacing, wrapping artifacts, unbalanced quotation marks, and fully capitalized formatting. This subsection describes adjacent-row merging to restore sentence boundaries.

Subtitle segmentation is then corrected when adjacent rows appear to belong to the same sentence. A continuation cue is evaluated separately for the EP and BP sentences after formatting normalization. It holds when the current row ends in soft punctuation, comprising a comma, semicolon, colon, em dash, en dash, or hyphen, or contains fewer than 40 characters, and the first alphabetic character of the following row is lowercase. A merge is allowed when this cue holds for either sentence.

A separate segmentation-mismatch cue applies when exactly one side of the current pair contains an internal hard stop followed later by another hard stop. This pattern suggests that material corresponding to more than one subtitle sentence may have been grouped on only one side. Hard punctuation comprises periods, question marks, exclamation marks, and ellipses. Rows are otherwise kept separate when both current sentences end in hard punctuation and the first alphabetic character of both following sentences is uppercase. The segmentation-mismatch cue overrides this safeguard.

Whenever a merge is triggered, both paired sentences are concatenated so that their row correspondence is preserved. The goal is to recover sentence boundaries split inconsistently by subtitle timing rather than to create longer examples indiscriminately.

Table 4.2 shows examples of these repairs.

A.2.2 SimAlign Configuration

After local cleanup, OpenSubtitles pairs are filtered with SimAlign (Jalili Sabet et al., 2020). SimAlign provides token-level links between the BP and EP sentences using contextual multilingual representations. This is useful for EP/BP data, since valid correspondences are not always surface-identical. A BP gerund may correspond to an EP infinitival construction, and a lexical item may correspond to a short phrase.

SimAlign requires three main choices. The first is the multilingual encoder used to compute contextual embeddings. The second is the granularity of the alignment, since links can be produced over subword pieces or over the original words after combining the representations of their subword pieces. The third is the extraction method that converts token similarities into alignment links. This work uses the XLM-R-based encoder configuration, word-level alignments, and the conservative intersection method, referred to as `inter` in SimAlign.

The extraction method operates over a similarity matrix, where rows correspond to tokens in the BP sentence, columns correspond to tokens in the EP sentence, and each cell estimates how close the two contextual representations are. The intersection method links two tokens only when each one is the other's highest-scoring option in this matrix. Two alternative SimAlign methods were also considered. The iterative maximum method begins from conservative mutual-best links, then iteratively adds further high-scoring links to cover tokens that remain unaligned, and can therefore produce more links than the intersection method. Maximum-weight matching considers the similarity matrix globally and selects a high-scoring one-to-one structure that maximizes the overall alignment score, rather than requiring every selected pair to be mutual local maxima. The intersection method was selected for filtering since it provides a conservative core of bilingual evidence and avoids accepting pairs through many weaker links.

Example. Consider BP tokens *[Ele] [está] [correndo]* and EP tokens *[Ele] [está] [a] [correr]*. The BP gerund *correndo* corresponds to the EP construction *a correr*. The following illustrative matrix shows the kind of token-similarity evidence used by the extraction step.

	Ele	*está*	*a*	*correr*
Ele	**0.94**	0.21	0.08	0.12
está	0.19	**0.91**	0.42	0.28
correndo	0.14	0.35	0.48	**0.86**

The selected mutual-best links are *Ele* to *Ele* (**0.94**), *está* to *está* (**0.91**), and *correndo* to *correr* (**0.86**). Each score is the highest-scoring option for both linked tokens. The *correndo* to *a* link (**0.48**) is not selected: *a* may prefer *correndo*, but *correndo* prefers *correr* at **0.86**, so the pair is not mutual-best. Intersection therefore captures the main gerund-to-infinitive correspondence *correndo* to *correr* while leaving the EP support token *a* unaligned.

A.2.3 Coverage Score and Length Filtering

To decide whether aligned pairs are kept in the final dataset, the alignment links provided by SimAlign are converted into a coverage score over normalized word tokens. Text is normalized with canonical Unicode normalization and case folding. The scoring tokens keep alphabetic forms, Portuguese diacritics, internal apostrophes or hyphens, and digit tokens. A token type is a distinct normalized token form, regardless of how many times that form occurs in the sentence. The coverage score is therefore type-based: repeated occurrences of the same normalized form do not contribute additional distinct types.

Let T_{ep} and T_{bp} be the resulting EP and BP token lists, respectively. Let V_{ep} and V_{bp} denote their sets of normalized token types, let A be the selected set of word-alignment pairs, and let V_A be the subset of token types that participate in at least one pair in A. The filter uses the following coverage score:

$$s(T_{\mathrm{bp}}, T_{\mathrm{ep}}) = \frac{|V_A|}{|V_{\mathrm{bp}} \cup V_{\mathrm{ep}}|}, \tag{A.1}$$

where the notation follows the definitions above.

The coverage score is then compared with a length-adaptive threshold. If $L = \max(|T_{\mathrm{bp}}|, |T_{\mathrm{ep}}|)$, the threshold is

$$\tau(L) = \begin{cases} 0.30, & L \leq 8, \\ 0.70, & L \geq 28, \\ 0.30 + \dfrac{L-8}{28-8}(0.70 - 0.30), & \text{otherwise.} \end{cases} \tag{A.2}$$

Short subtitle fragments are allowed lower coverage, while longer pairs must show stronger bilingual alignment. A separate length-ratio check rejects pairs where one sentence is implausibly short or long relative to the other. If r is the ratio between EP and BP character length, the pair is kept only when $0.5 \leq r \leq 2.0$. The final rule retains pairs meeting alignment-coverage and length-ratio constraints.

A.3 GPT-Wikipedia

GPT-Wikipedia is an internally generated resource of near-literal EP/BP pairs whose sentences are generated through the GPT API using Portuguese Wikipedia passages as inspiration. After generation, each pair passes through an automatic audit before human review. The EP and BP sentences are first tokenized, then the tokens are normalized by lowercasing, removing diacritics, and suppressing hyphen differences. The two ordered token sequences are compared so that matching tokens or regions are aligned, while contiguous insertions, deletions, or replacements become differing spans.

The automatic audit then checks these differing spans against a small allowlist of expected EP/BP correspondences. Remaining differences are flagged for review and assigned one of three reason categories: unexpected or non-approved lexical replacements, structural or word-order differences, and reordering artifacts. Repeated normalized replacements are grouped together and ordered by frequency. If the same EP expression and BP expression occur in several generated pairs, the reviewer sees that expression pair once together with its occurrence count rather than reviewing every sentence pair separately.

Human review labels each grouped replacement as either a valid variety contrast or an unnecessary change, and high-frequency replacements can be reviewed in batches. The review scripts can propagate decisions conservatively to related inflectional forms and containing phrases. When the resulting decision is applied to matching generated examples, a valid variety contrast preserves the EP form on the EP side and the BP form on the BP side, whereas an unnecessary change is equalized across the pair. This applies the recorded and conservatively propagated decisions consistently without requiring pair-by-pair review of the complete resource.

B

LoRA Rank Ablation

This appendix reports a LoRA-rank ablation for the decoder-controlled 4B-parameter T5Gemma 2 model. The comparison examines how performance changes across LoRA ranks under the supervised training setup used for the corresponding Stage B GPT-Wikipedia configuration.

B.1 Ablation Design

The 4B-parameter experiments use LoRA rank 48. This ablation compares ranks 8, 16, 32, and 48 in the decoder-controlled configuration trained with Stage A on OpenSubtitles followed by Stage B on GPT-Wikipedia. All runs use the supervised protocol selected in Section 6.2. The control strings, data splits, target modules, learning rates, dropout, and evaluation protocol are kept fixed, together with the LoRA scaling factor $\alpha = 96$.

However, the implementation scales the LoRA update by α/r, so holding α fixed means that the effective adapter multiplier changes with rank. The resulting multipliers are 12, 6, 3, and 2 for ranks 8, 16, 32, and 48, respectively. The comparison therefore varies rank together with this effective scaling and should not be interpreted as an isolated effect of rank alone. Table B.1 reports the results on the Golden Collection dataset, as it is the benchmark most closely aligned with the conservative rewriting objective.

B.2 Results

Rank 48 gives the best observed rewriting performance, with the highest BLEU (87.68) and lowest TER (0.0895). However, its BLEU advantage is modest: 0.44 points over rank 16, the next-highest result, and 0.54 points over rank 8. Ranks 8–32 are tightly clustered between 87.14 and 87.24 BLEU, while rank 8 uses approximately one-sixth as many trainable adapter parameters as rank 48. Identification does not follow the same ordering: rank 16 gives the highest macro-F1, at 72.31 compared with 71.53 for rank 48,

Table B.1: Decoder-controlled LoRA rank ablation on the Golden Collection.

Rank	Trainable parameters		Macro-F1 (%)	BLEU (0–100)	TER
	Count	% of rank 48			
8	31.3M	16.7%	70.95	87.14	0.0943
16	62.6M	33.3%	**72.31**	87.24	0.0964
32	125.2M	66.7%	71.17	87.23	0.0944
48	187.8M	100.0%	71.53	**87.68**	**0.0895**

and macro-F1 does not increase monotonically with rank. The results therefore show a trade-off between trainable adapter size and performance. Rank 48 performs best on rewriting in this configuration, while smaller adapters remain competitive when parameter efficiency matters.

The experiment establishes neither statistical significance nor rank-performance monotonicity.

Reducing LoRA rank lowers the number of trainable adapter parameters, but it does not reduce the size of the frozen base model or proportionally reduce activation and forward-pass costs. The percentages in Table B.1 describe adapter-parameter savings, not proportional savings in training memory or computation.

C

Isolated Reward-Based Continuation Ablations

This appendix reports isolated reward-design diagnostics on the fully fine-tuned 270M-parameter T5Gemma 2 model. They are separate from the Stage C continuations with the 4B-parameter model described in Sections 5.3.3 and 6.4. Four independent diagnostics examine the base normalized BLEU/TER reward and three modifications involving classifier feedback, token conservatism, and candidate-group construction. Both encoder-controlled and decoder-controlled formulations are examined.

C.1 Experimental Setup

Each diagnostic begins independently from the selected supervised Stage B checkpoint of the 270M-parameter model for its control formulation. These checkpoints follow Stage A adaptation on OpenSubtitles and Stage B specialization on GPT-Wikipedia under the supervised protocol selected in Section 6.2. No reward variant is continued from another.

The two control formulations start from different supervised checkpoints, so each diagnostic is interpreted relative to its own Stage B baseline rather than compared in absolute terms.

All continuations use the same 1200-step optimization budget. Evaluation follows the 270M-parameter protocol-selection comparison, reporting macro-F1 for identification and aggregate corpus BLEU and TER for rewriting on FRMT and the Golden Collection.

C.2 Reward Variants

The base reward uses the normalized BLEU and TER components defined in Section 5.3.3. For each generated candidate, sentence-level BLEU and TER are computed against the reference rewrite and normalized relative to the scores obtained by leaving the source unchanged. The base reward weights both components equally and penalizes source copies by 0.10 when the reference requires a change.

In the decoder-controlled diagnostic, the generated `PT/BR` control label is also checked before the sentence-level reward is used. An incorrect first label sets the candidate reward to zero, whereas main decoder-controlled Stage C uses a weighted first-label component.

The classifier-reward variant adds an auxiliary variety-identification score. Each generated rewrite is passed back through the model using the identification formulation for the same control setup, and the probability assigned to the intended variety of the rewritten sentence provides the classifier score. This measures whether the generated sentence itself exhibits the requested variety. Under decoder control, this component is separate from the binary first-label gate, which checks only the generated control token.

The token-conservatism variant adds a token-level score. After lowercasing and tokenizing the source, candidate, and reference, each generated token receives one of three scores. A token absent from the reference receives 0. A token present in both the reference and source receives 0.5. A token present in the reference but absent from the source receives 1.0. The sequence score averages these token scores.

For both auxiliary-reward variants, normalized BLEU and TER each receive a weight of 0.45, while the added component receives a weight of 0.10.

The TER-diverse grouping variant changes candidate-group construction rather than the reward formula. For each input, a pool of 12 candidates is first sampled and their TER against the reference is computed. After duplicate normalized candidate texts are removed, the remaining candidates are ordered by TER and up to four are selected at evenly spaced quantile positions across that ordering, producing a group that covers different levels of distance from the reference. The reference candidate is then appended, so the group-relative optimization normally compares four selected samples with the reference candidate. The selected candidates use the same normalized BLEU/TER reward as the base configuration. This diagnostic therefore changes the examples compared within the group-relative update rather than adding another reward component.

C.3 Results

Tables C.1 and C.2 report the decoder-controlled and encoder-controlled diagnostics, respectively.

For the decoder-controlled diagnostics, FRMT rewriting remains close to the supervised Stage B starting point across all four variants, with a largest absolute BLEU change of only 0.09 points. However, on the Golden Collection, TER-diverse grouping gives the best rewriting result among the diagnostics, reaching 87.19 BLEU and 0.0949 TER compared with 86.57 BLEU and 0.0980 TER at the supervised starting point. Its macro-F1 is lower, at 66.70 compared with 67.63. Token conservatism does not improve Golden Collection TER, reaching 0.1007 compared with 0.0980 at the starting point.

Table C.1: Decoder-controlled 270M isolated Stage C ablations.

Configuration	FRMT			Golden Collection		
	Macro-F1 (%)	BLEU (0–100)	TER	Macro-F1 (%)	BLEU (0–100)	TER
Supervised Stage B start	**70.76**	**39.13**	**0.4788**	**67.63**	86.57	0.0980
Base reward	68.76	39.11	0.4800	67.15	86.88	0.0962
Classifier reward	69.54	39.04	0.4806	67.57	86.96	0.0969
Token conservatism	69.49	39.07	0.4804	67.43	86.58	0.1007
TER-diverse grouping	68.24	39.12	0.4799	66.70	**87.19**	**0.0949**

Table C.2: Encoder-controlled 270M isolated Stage C ablations.

Configuration	FRMT			Golden Collection		
	Macro-F1 (%)	BLEU (0–100)	TER	Macro-F1 (%)	BLEU (0–100)	TER
Supervised Stage B start	71.73	**27.06**	**0.6204**	65.43	**65.53**	**0.2996**
Base reward	71.90	13.62	0.7525	65.77	38.21	0.5154
Classifier reward	71.81	15.67	0.7319	66.29	44.25	0.4704
Token conservatism	**72.28**	16.27	0.7268	**66.34**	42.79	0.4808
TER-diverse grouping	72.22	13.18	0.7571	66.14	38.96	0.5118

The encoder-controlled diagnostics show a different pattern. Every reward-based continuation substantially reduces rewriting quality relative to the supervised Stage B starting point. On FRMT, the best diagnostic BLEU is 16.27 under token conservatism, compared with 27.06 at the starting point. On the Golden Collection, classifier reward gives the highest diagnostic BLEU at 44.25, compared with 65.53 before continuation. Identification changes are much smaller than the rewriting changes, with token conservatism giving the highest macro-F1 among the diagnostics on both FRMT and the Golden Collection.

Taken together, no tested reward modification improves rewriting consistently across both control formulations. Decoder-controlled continuation remains close to its supervised starting point and yields modest Golden Collection gains under some variants, with TER-diverse grouping giving the best decoder-controlled rewriting result. In contrast, every encoder-controlled continuation substantially reduces rewriting despite much smaller identification changes. The classifier and token-conservatism components provide no consistent advantage across formulations, so no tested modification is generally preferable.

D

External Model Comparisons

This appendix evaluates four external pretrained or instruction-following systems on the same EP/BP identification and rewriting tasks to provide context for the adapted T5Gemma 2 models evaluated in Chapter 6. The systems are AMALIA-9B-0626-SFT and AMALIA-9B-0626-DPO (Simplício et al., 2026), Qwen3-4B (Yang et al., 2025), and FLAN-T5-XL (Chung et al., 2024).

D.1 Evaluation Protocol

The evaluated checkpoints are `amalia-llm/AMALIA-9B-0626-SFT`, `amalia-llm/AMALIA-9B-0626-DPO`, `Qwen/Qwen3-4B`, and `google/flan-t5-xl`.

Rewriting prompts specify whether the sentence should be rewritten from BP to EP or from EP to BP. The generated sentences are evaluated with the same BLEU and TER scoring procedure used for the adapted models, with the two directions reported separately.

Identification is performed differently across the external systems. AMALIA SFT and AMALIA DPO generate an answer freely and the predicted variety label is then parsed from the output, whereas Qwen3-4B and FLAN-T5-XL score the possible natural-language labels directly and select the higher-scoring option. The comparison therefore reflects each system together with its identification inference procedure.

The systems differ in architecture, scale, training history, prompting, and inference procedure, so the comparison provides external context rather than a controlled model comparison.

D.2 Results

Table D.1 reports the external-model results on both evaluation resources.

AMALIA DPO gives the highest external identification macro-F1 on both evaluation resources, but remains below the best adapted configurations reported in Table 6.2.

Table D.1: External-model results on FRMT and the Golden Collection.

Model	Dataset	Macro-F1 (%)	BP-to-EP BLEU (0–100) / TER	EP-to-BP BLEU (0–100) / TER
AMALIA SFT	FRMT	45.37	**44.65**/**0.4289**	40.22/0.4682
	Golden Collection	43.99	68.44/0.2347	74.66/0.1920
AMALIA DPO	FRMT	**59.72**	43.82/0.4494	35.16/0.6063
	Golden Collection	**56.55**	58.08/0.3239	57.18/0.3773
Qwen3-4B	FRMT	52.56	39.44/0.4709	**40.73**/**0.4654**
	Golden Collection	49.96	**78.33**/**0.1540**	**78.26**/**0.1590**
FLAN-T5-XL	FRMT	47.87	32.22/0.5263	32.17/0.5373
	Golden Collection	40.99	67.06/0.2209	66.31/0.2292

On the Golden Collection, Qwen3-4B performs best among the external rewriting systems in both directions, but the adapted configurations perform better for both BP-to-EP and EP-to-BP in Table 6.3.

On FRMT, AMALIA SFT slightly outperforms the best adapted BP-to-EP result, with 44.65 BLEU compared with 43.97, while Qwen3-4B remains below the best adapted EP-to-BP result.

Overall, the adapted models perform better on identification and Golden Collection rewriting, while the external systems are competitive on FRMT only in the BP-to-EP direction.

E

Secondary No-Equal Ablation

The decoder-controlled no-equal variant removes equal source-reference rewriting examples from the supervised stages while retaining identification examples. It uses the same Stage C continuation data as the matched configuration. Stage C therefore differs through its supervised starting checkpoint rather than through a separate continuation-data variant.

Table E.1 reports the aggregate results for this ablation.

Table E.1: Secondary decoder-controlled no-equal ablation under aggregate evaluation.

Stage	Data condition	Dataset	Accuracy (%)	Macro-F1 (%)	BLEU (0–100)	TER
Stage B	GPT-Wikipedia	FRMT	81.98	81.86	41.31	0.4605
		Golden Collection	71.61	71.38	86.29	0.0981
Stage B	GPT-Wikipedia + FRMT	FRMT	**86.84**	**86.84**	**44.73**	**0.4297**
		Golden Collection	71.43	71.37	62.52	0.2833
Stage C	GPT-Wikipedia	FRMT	82.45	82.37	40.80	0.4643
		Golden Collection	**73.08**	**73.00**	**87.79**	**0.0889**
Stage C	GPT-Wikipedia + FRMT	FRMT	85.86	85.78	43.53	0.4409
		Golden Collection	70.70	70.47	80.03	0.1468

Removing equal rewriting examples produces only small changes on FRMT but generally reduces Golden Collection rewriting performance, supporting retention of equal examples in the supervised data. The reduction is consistent with the copy-friendly structure of the Golden Collection discussed in Section 6.5.1, where many examples require little or no rewriting.